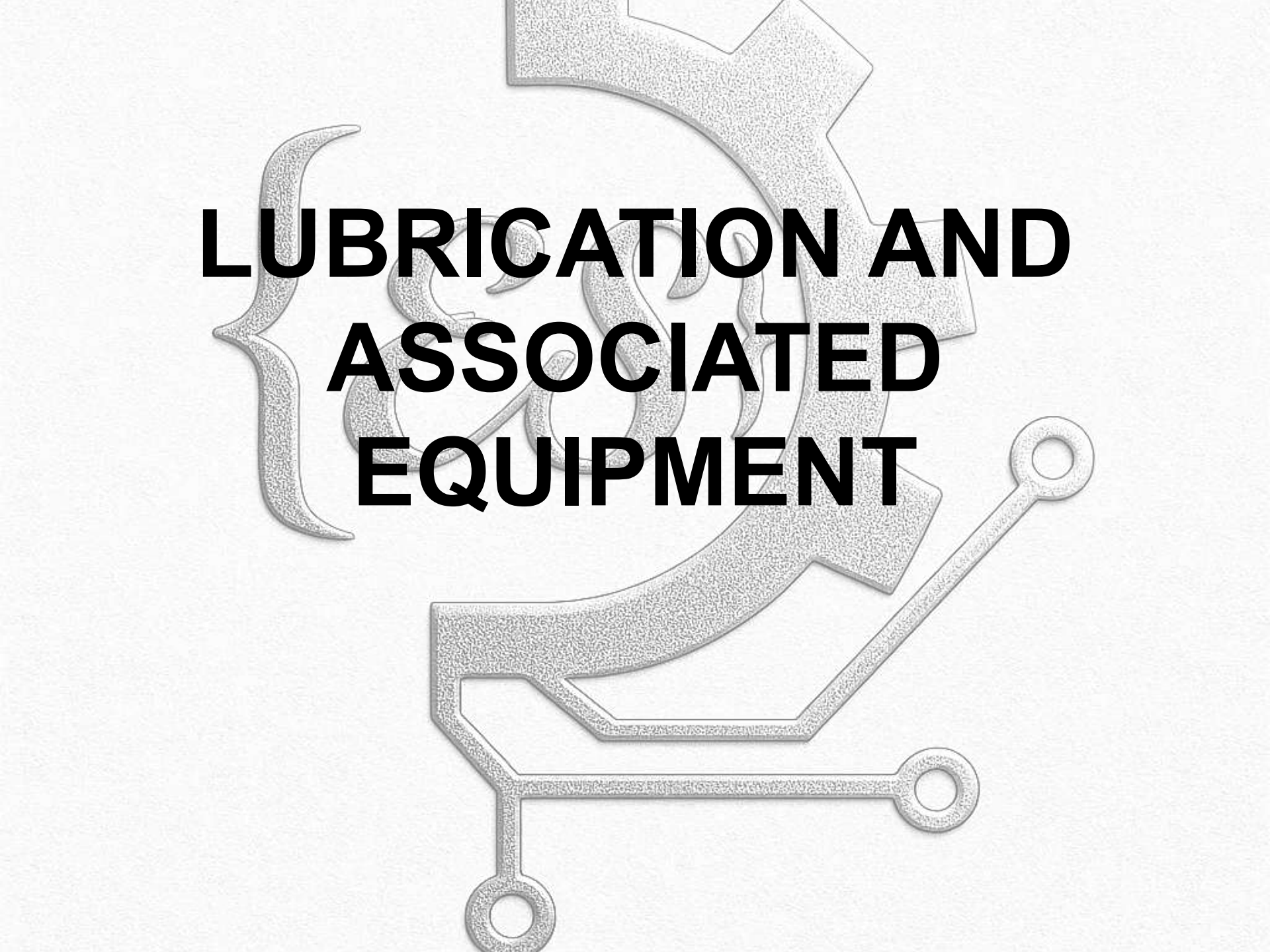


The background of the slide features a large, stylized gear and a mechanical arm. The gear is positioned in the upper left and center, while the mechanical arm, which has three circular joints, extends from the bottom left towards the center. The entire graphic is rendered in a light gray, textured style.

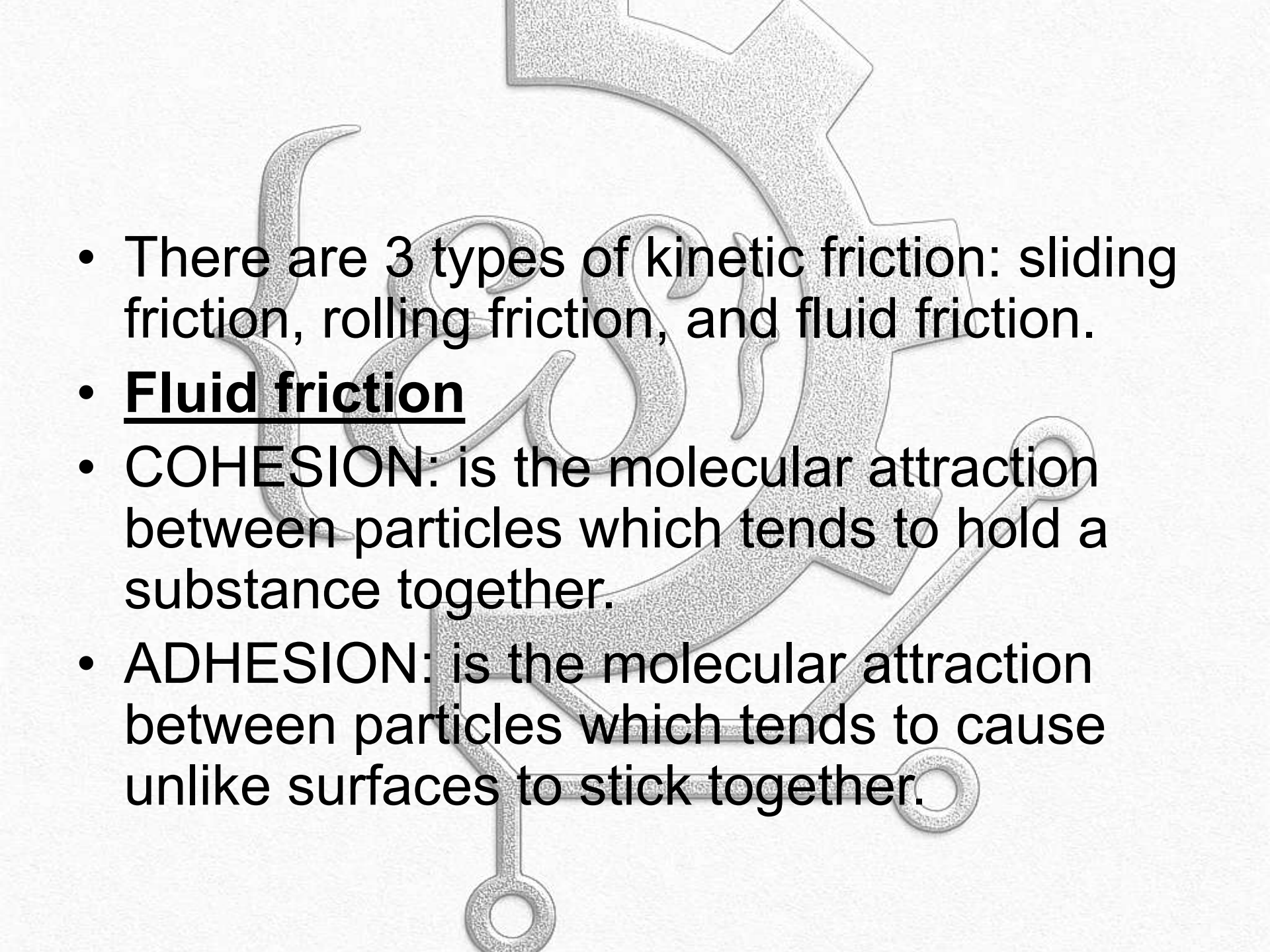
LUBRICATION AND ASSOCIATED EQUIPMENT

References:

- Ship's Information Book (SIB), Vol. 2
- Propulsion Plant Manual (PPM)
- NAVEDTRA 14151, Machinist's Mate 3 & 2 (Surface)
- Naval Ship's Technical Manual (NSTM) 262.

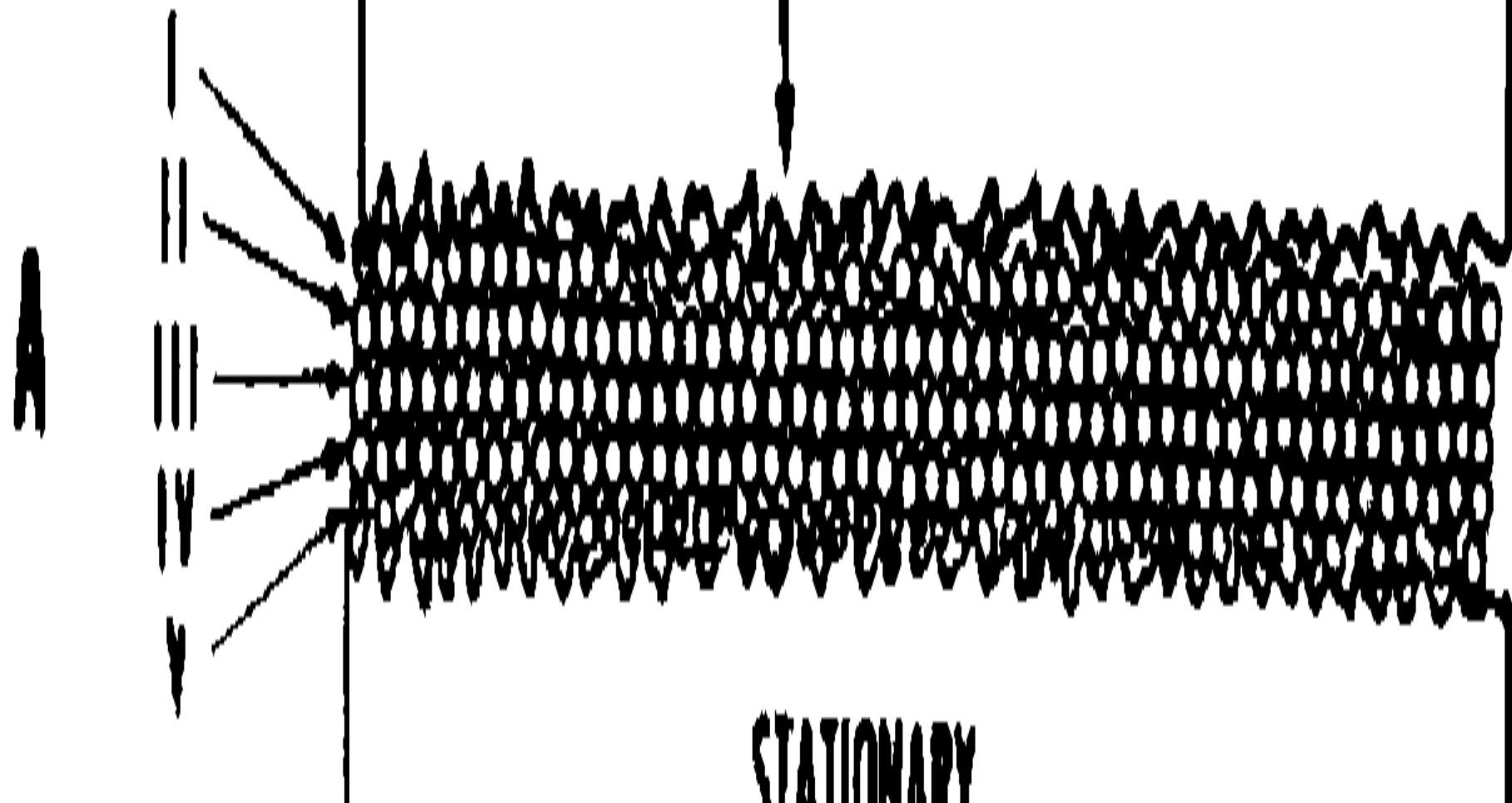
SYSTEM COMPONENTS AND COMPONENT PARTS

- *The friction that exists between a body at rest and the surface upon which it rests is called **STATIC** friction.*
- *The friction that exists between moving bodies (or between one moving body and a stationary surface) is called **KINETIC** friction.*
- *Static friction is greater than kinetic friction.*

- 
- There are 3 types of kinetic friction: sliding friction, rolling friction, and fluid friction.
 - **Fluid friction**
 - **COHESION:** is the molecular attraction between particles which tends to hold a substance together.
 - **ADHESION:** is the molecular attraction between particles which tends to cause unlike surfaces to stick together.

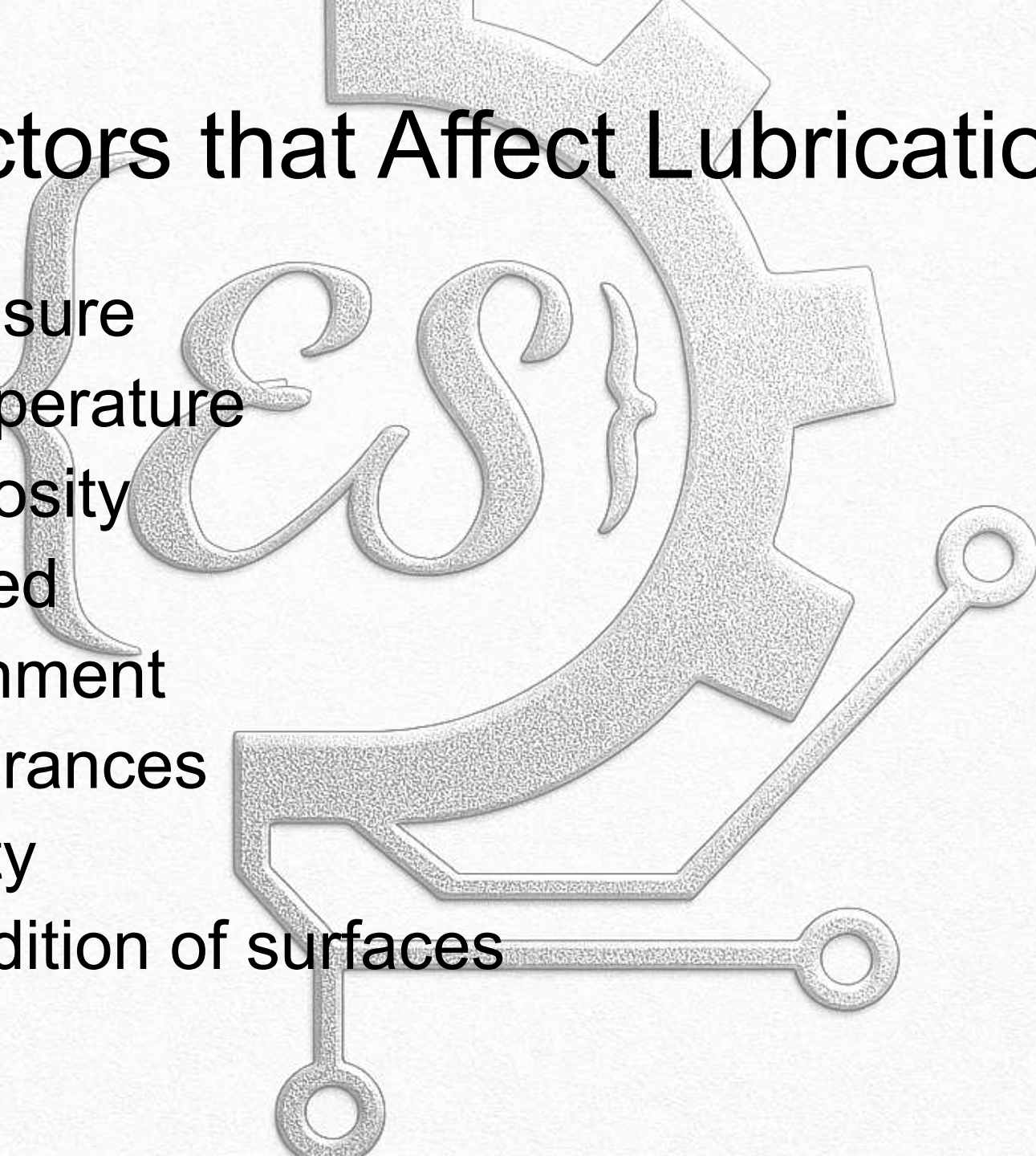
LANGMUIR THEORY

- The theory of lubrication is based on the Langmuir theory. Theoretically, three or more layers or films of oil exist between two lubricated bearing surfaces. Two of the films are BOUNDARY films (indicated as I&V), one of which clings to the surface. When two metal surfaces are in motion, oil film II, III, and IV creates a sliding wedge that prevents direct contact.



Factors that Affect Lubrication

- Pressure
- Temperature
- Viscosity
- Speed
- Alignment
- Clearances
- Purity
- Condition of surfaces



Classification of Lubricating Oils

2190 TEP

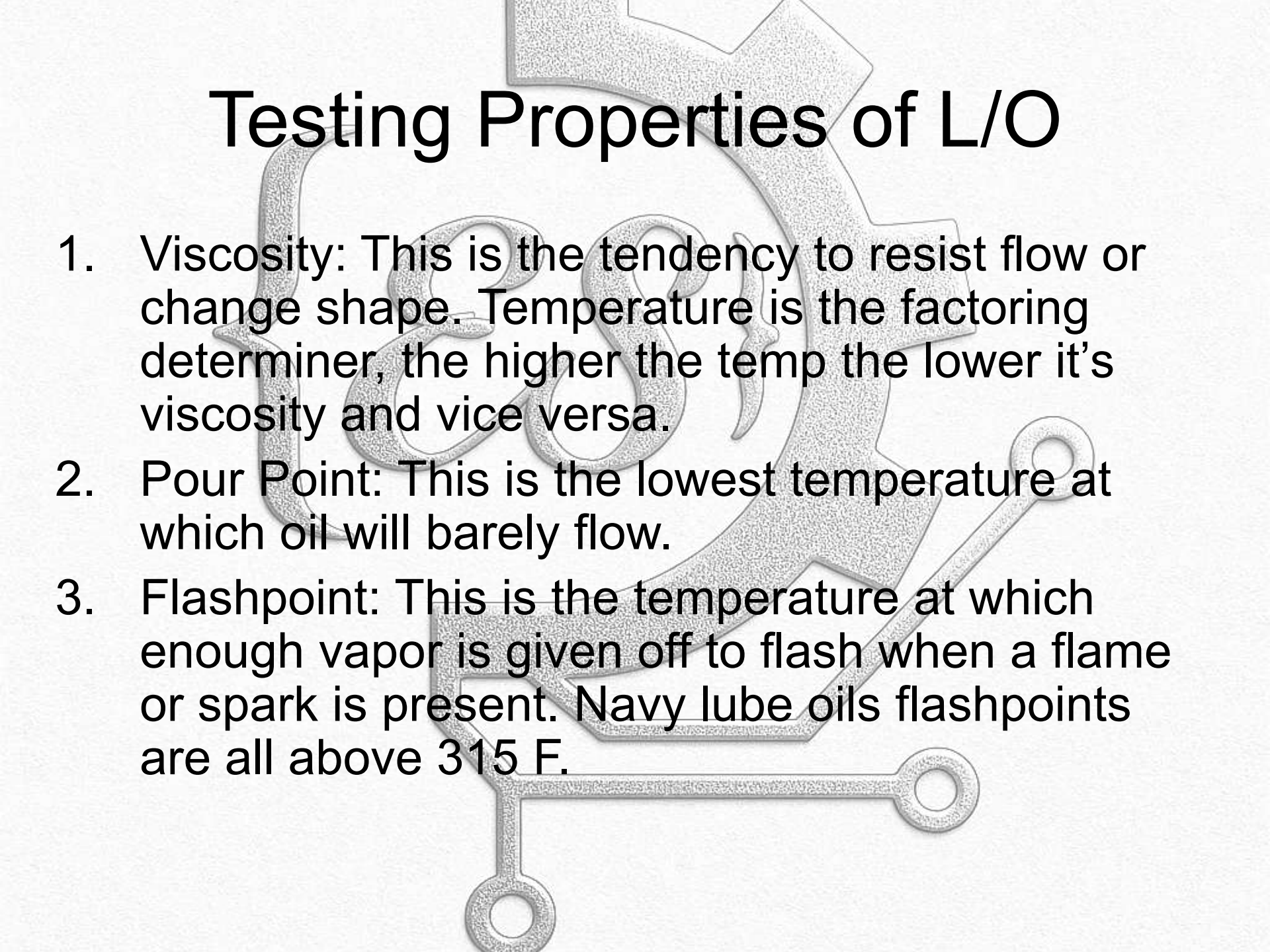
Series

i.e. 2000 series

**Special Additives (for
Turbine Extreme Pressure)**

***60 ml Sample Flows Through a
Universal Saybolt Orifice in 190
Seconds at 130F***

Testing Properties of L/O

A large, light gray gear is centered in the background. Overlaid on the gear is a dipstick, also in light gray, with a circular handle at the top and a rectangular base at the bottom.

1. Viscosity: This is the tendency to resist flow or change shape. Temperature is the factoring determiner, the higher the temp the lower it's viscosity and vice versa.
2. Pour Point: This is the lowest temperature at which oil will barely flow.
3. Flashpoint: This is the temperature at which enough vapor is given off to flash when a flame or spark is present. Navy lube oils flashpoints are all above 315 F.

Testing Properties of L/O

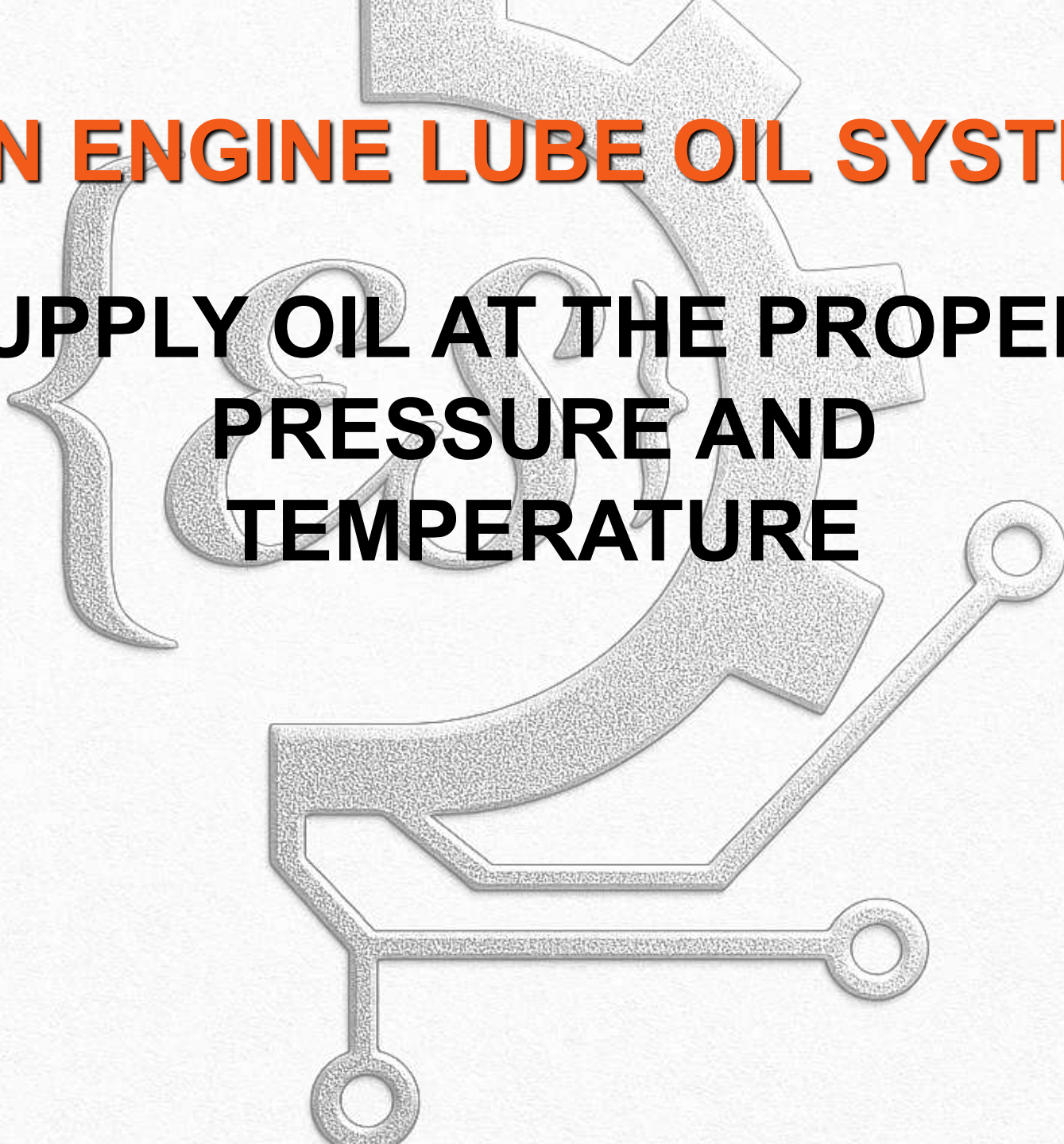
4. Fire Point: This is the temperature at which the oil will continue to burn when ignited.
5. Auto-Ignition Point: This is the point at which flammable vapors given off from oil will burn.
6. Demulsibility: The ability of oil to separate cleanly from any water present.

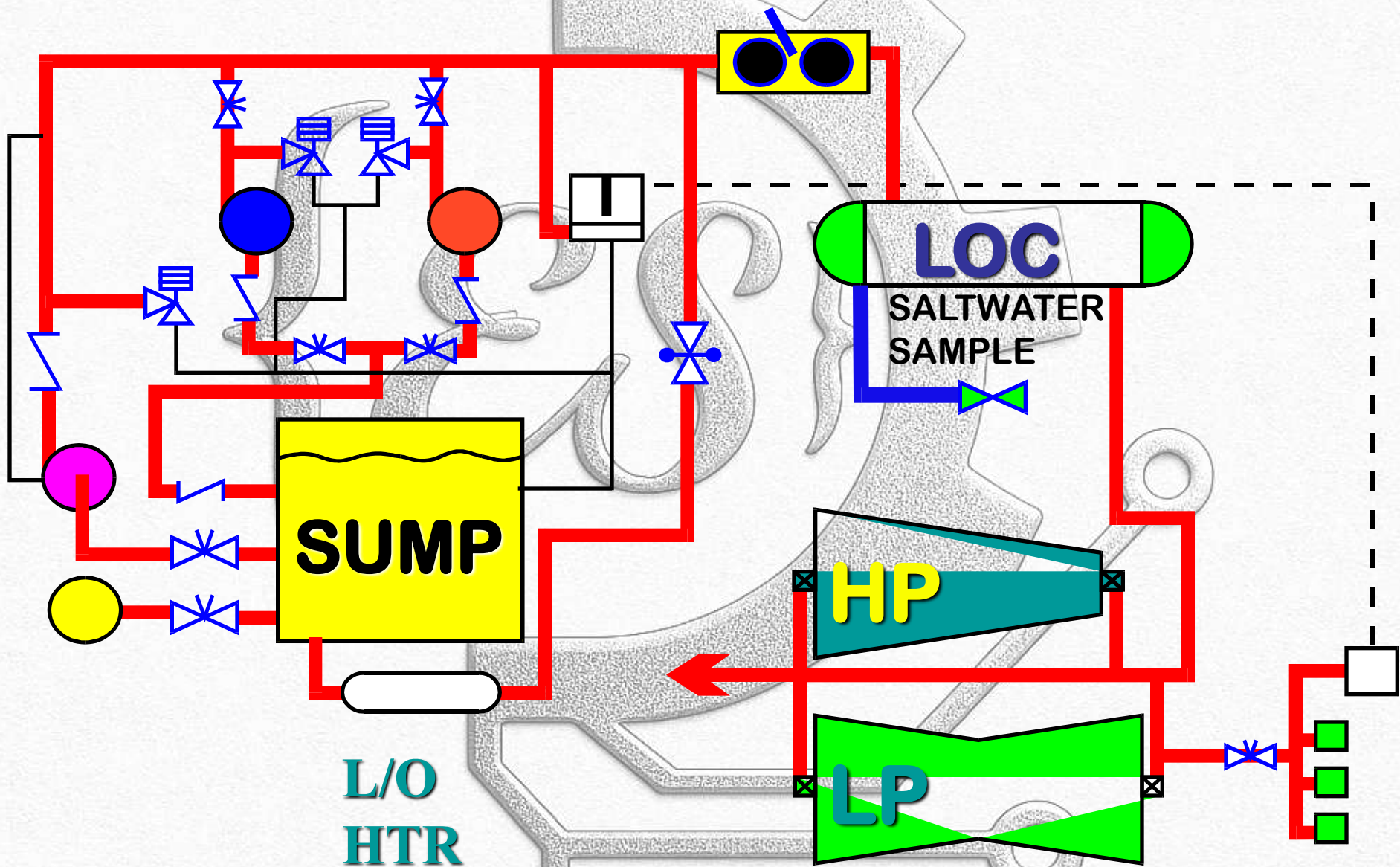
Testing Properties of L/O

7. Neutralization Number: This is the measure of acid content in oil.
8. Precipitation Number: The measure of the amount of solids classified as asphalts or carbon residue in oil.

MAIN ENGINE LUBE OIL SYSTEM

**SUPPLY OIL AT THE PROPER
PRESSURE AND
TEMPERATURE**





MAIN LUBE OIL SYSTEM

PRINCIPLES OF OPERATION

- *How do the components work together to achieve the system's function?*

Normally, the attached main lubricating oil service pump supplies oil to all bearings at a minimum pressure of 22 psig. When the oil pressure at the most remote bearing drops to 16 psig, the stand-by pump will speed up automatically and bring the oil pressure back to 20-22 psig.

PARAMETERS/OPERATING LIMITS

- ***Sump level-*** 1800 gallons
- ***Lube oil pump discharge pressure***
 - Attached lubricating oil pump discharge Pressure 0-60 psig
 - Turbine-driven lubricating oil pump discharge pressure 0-60 psig
 - Motor-driven lubricating oil pump discharge pressure 0-60 psig

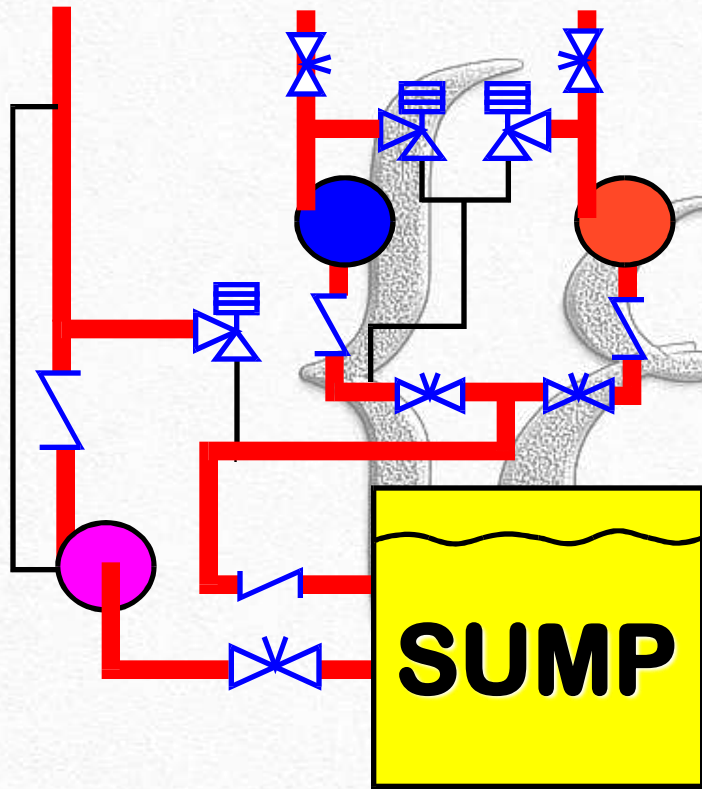
PARAMETERS/OPERATING LIMITS

- ***Lube oil pressure at remote bearing-***
 - 22 psig
- ***Lube oil strainer differential pressure***
 - Outlet no more than 7.5 psig below inlet
- ***Lube oil cooler outlet temperature***
 - 120°-130°F

PARAMETERS/OPERATING LIMITS

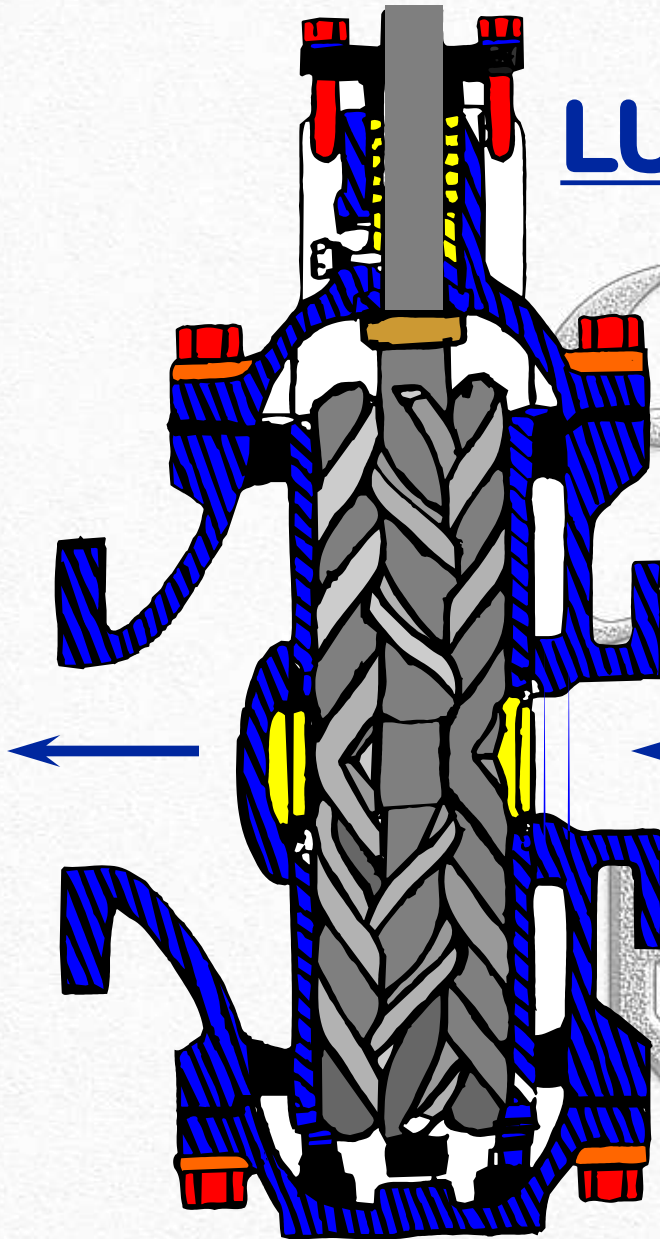
- *Bearing lube oil outlet temperature*
 - 140°F
- *Motor-driven pump cut-in switch*
 - 16 psig cut-in manual cut-out
- *Lubricating oil low pressure alarm*
 - 9 psig cut-in 10-11 psig cut-out

Lube Oil Pumps



LUBE OIL SERVICE PUMP

- Triple Screw
- Positive displacement
- Steam or Electric driven

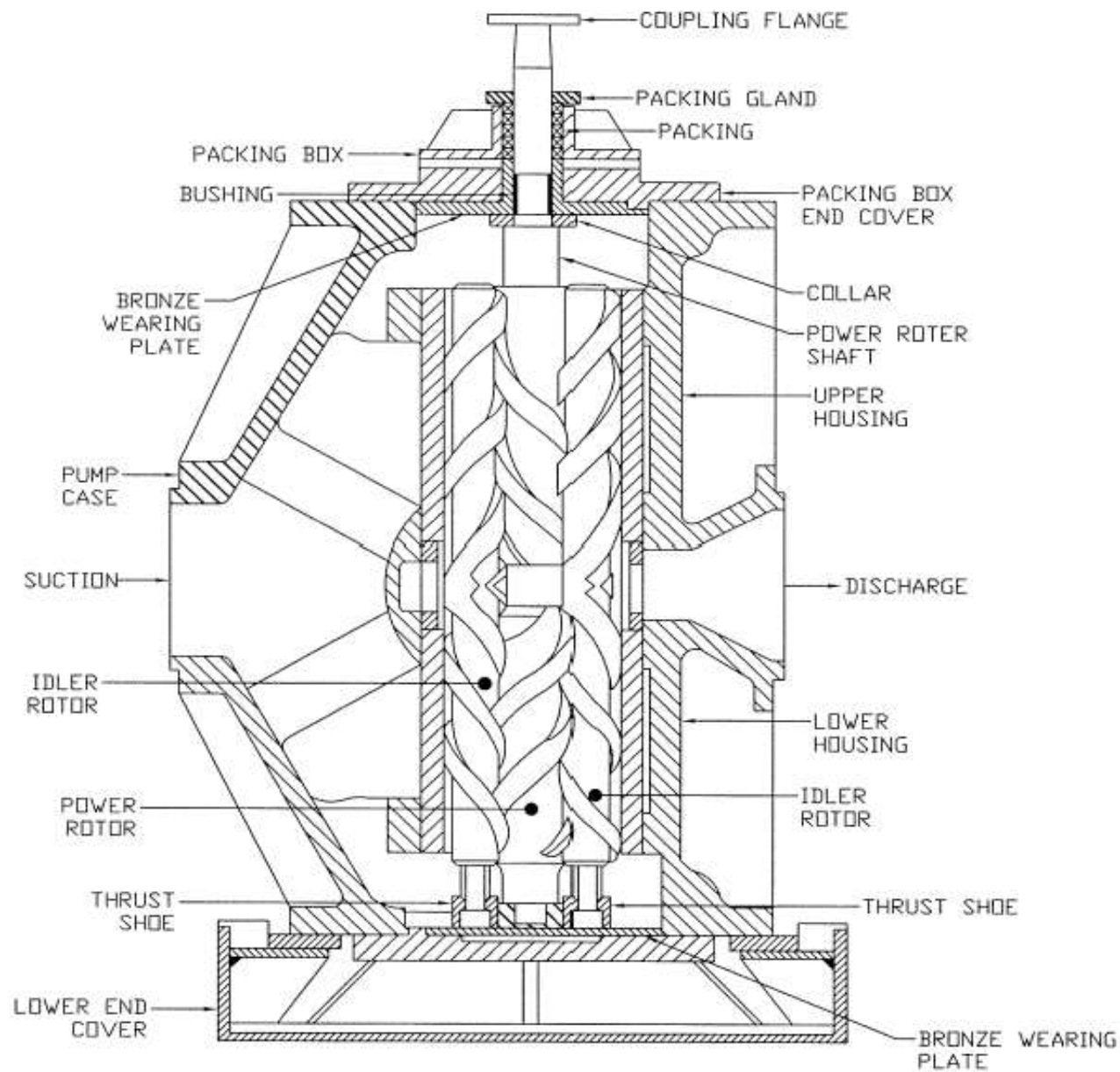


SYSTEM COMPONENTS AND COMPONENT PARTS (cont'd)

- **SLOP-** This turbine-driven, rotary, screw-type, vertical, positive displacement pump has a power rotor that propels the oil, two idler rotors for sealing the oil, upper and lower housings, shaft packing, and piping connections. Idler rotors are driven by the power rotor with which they mesh.

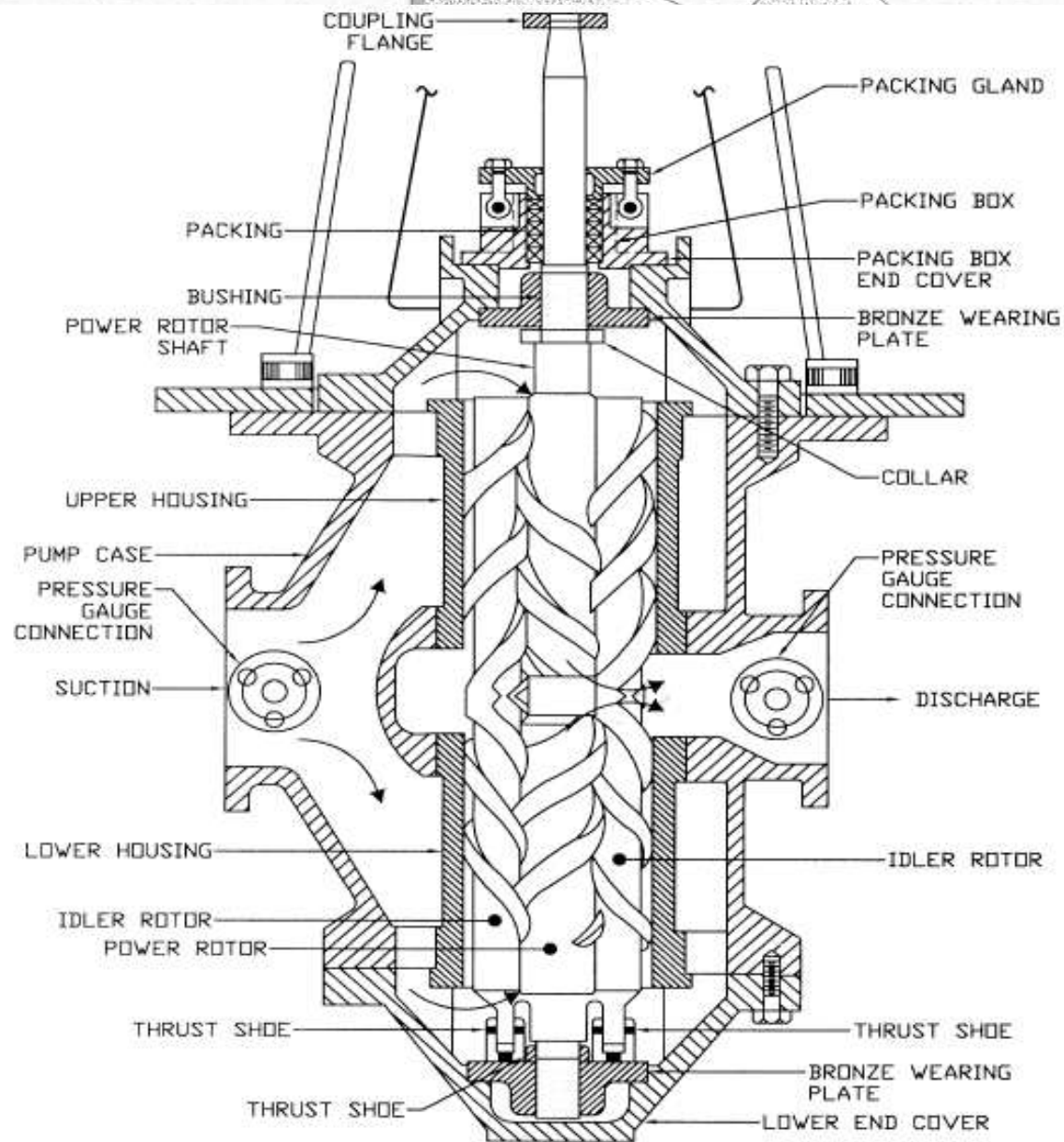
SLOP (cont'd)

- All three rotors are enclosed in a close fitting housing, and the intermeshing threadlike forms of the rotors are shaped so that leakage from the discharge passage to the suction passage is negligible at all operating pressures. The power rotor carries the entire pumping load, while the idler rotors simply act as seals.



SYSTEM COMPONENTS AND COMPONENT PARTS (cont'd)

- **ELOP.** This is a motordriven, rotary, screw-type, vertical, positive displacement pump. Construction and component nomenclature are basically the same as that of the standby main lubricating oil service pump.



SYSTEM COMPONENTS AND COMPONENT PARTS (cont'd)

- **ALOP-** This is a General Electric, rotary gear, vertically mounted positive displacement pump that consists of a three-gear element assembled and supported in a cast bronze housing. The pump is driven from the forward end of the second reduction gear shaft through a flexible coupling.

ALOP (cont'd)

- The pump is arranged in two main parts: the drive, supported by the forward end of the lower gear casing; and the pump assembly (below the drive assembly), supported by the main gear foundation. The oil pump drive is driven from the forward end of the second reduction gear through a coupling sleeve mounted at the center of the second reduction gear shaft. Power is delivered through a coupling distance piece to a coupling sleeve that is bolted directly to the main drive gear.

SEQUENCE OF EVENTS(SEQUENCING)

TWO ELECTRIC PUMPS

- ▲ **15 PSIG- NORMAL PRESSURE.**
- ▲ **12 PSIG- ON-LINE PUMP → HIGH SPEED.**
- ▲ **11 PSIG- EMERGENCY PUMP START IN LOW SPEED.**
- ▲ **9 PSIG - EMERGENCY PUMP → HIGH SPEED.**
- ▲ **7 PSIG - LUBE ALARM WILL SOUND.**

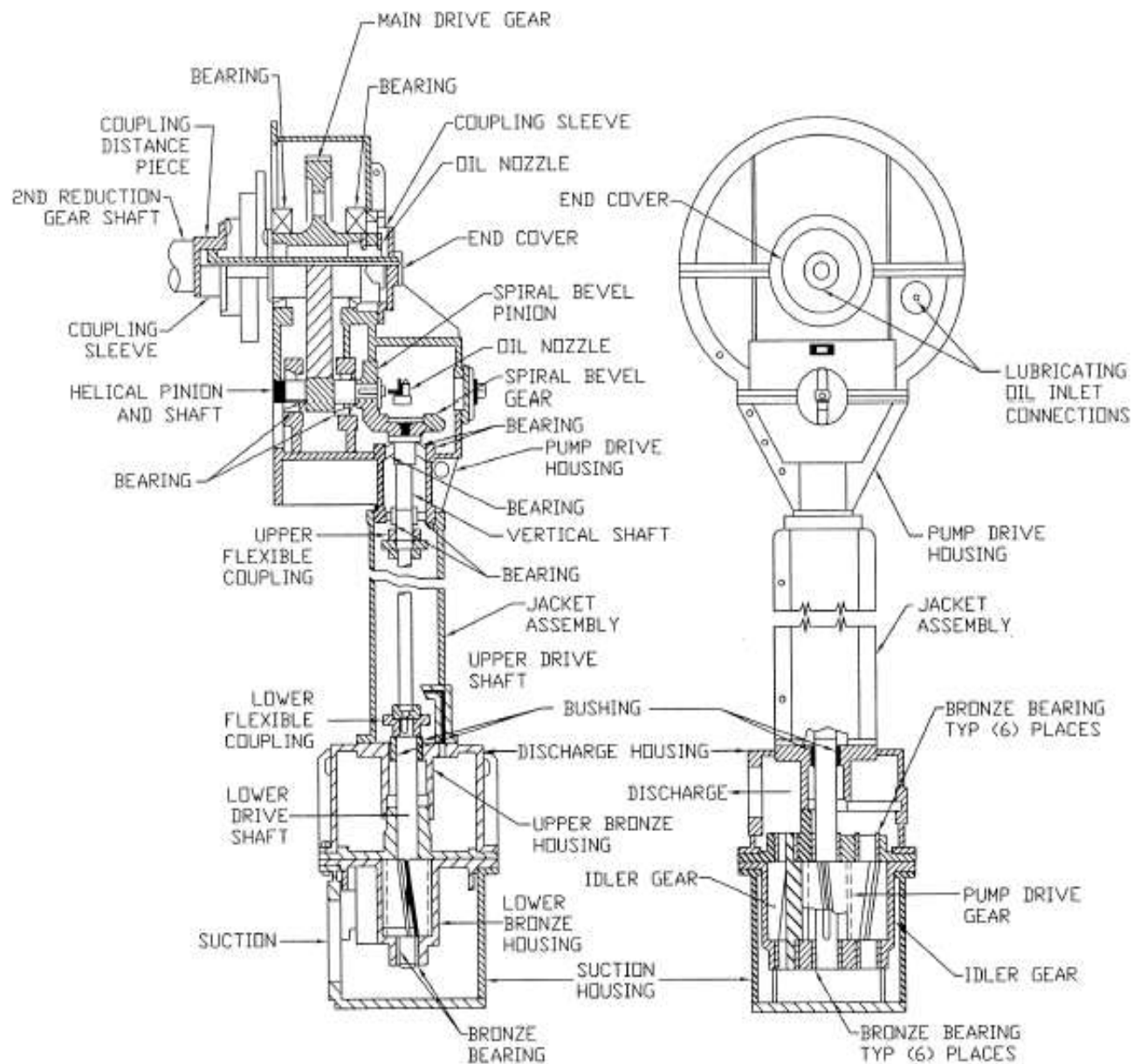


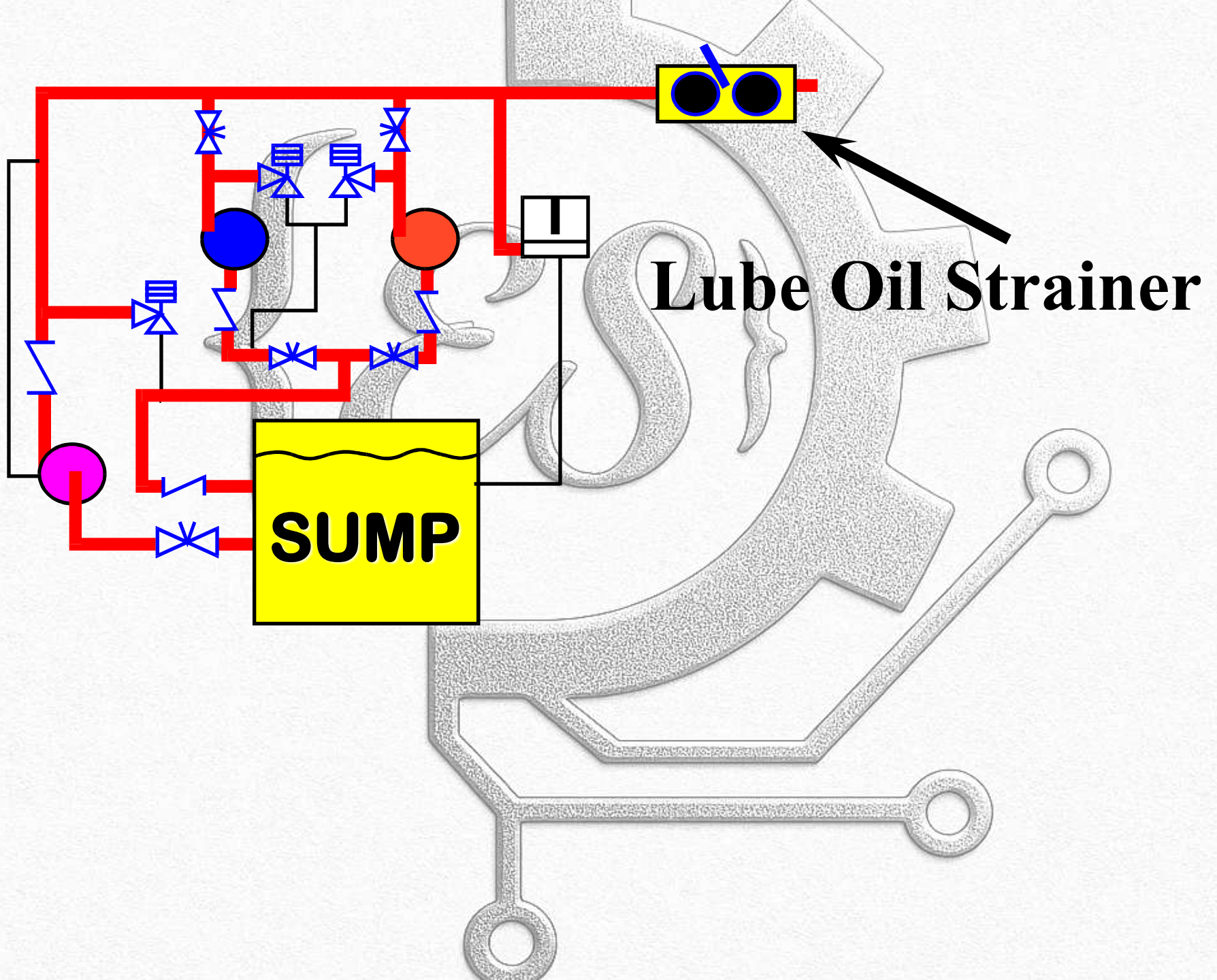
SEQUENCE OF EVENTS

ONE STEAM/ONE ELECTRIC

- ▶ 15 PSIG- NORMAL PRESSURE.
- ▶ AS PRESSURE DROPS THE UNLOADER WILL SHUT.
- ▶ IF PRESSURE CONTINUES TO DROP THE STEAM ADMISSION VALVE WILL OPEN TO SPEED UP THE PUMP.
- ▶ 11 PSIG- EMERGENCY PUMP STARTS.
- ▶ 7 PSIG- LUBE OIL ALARM WILL SOUND.



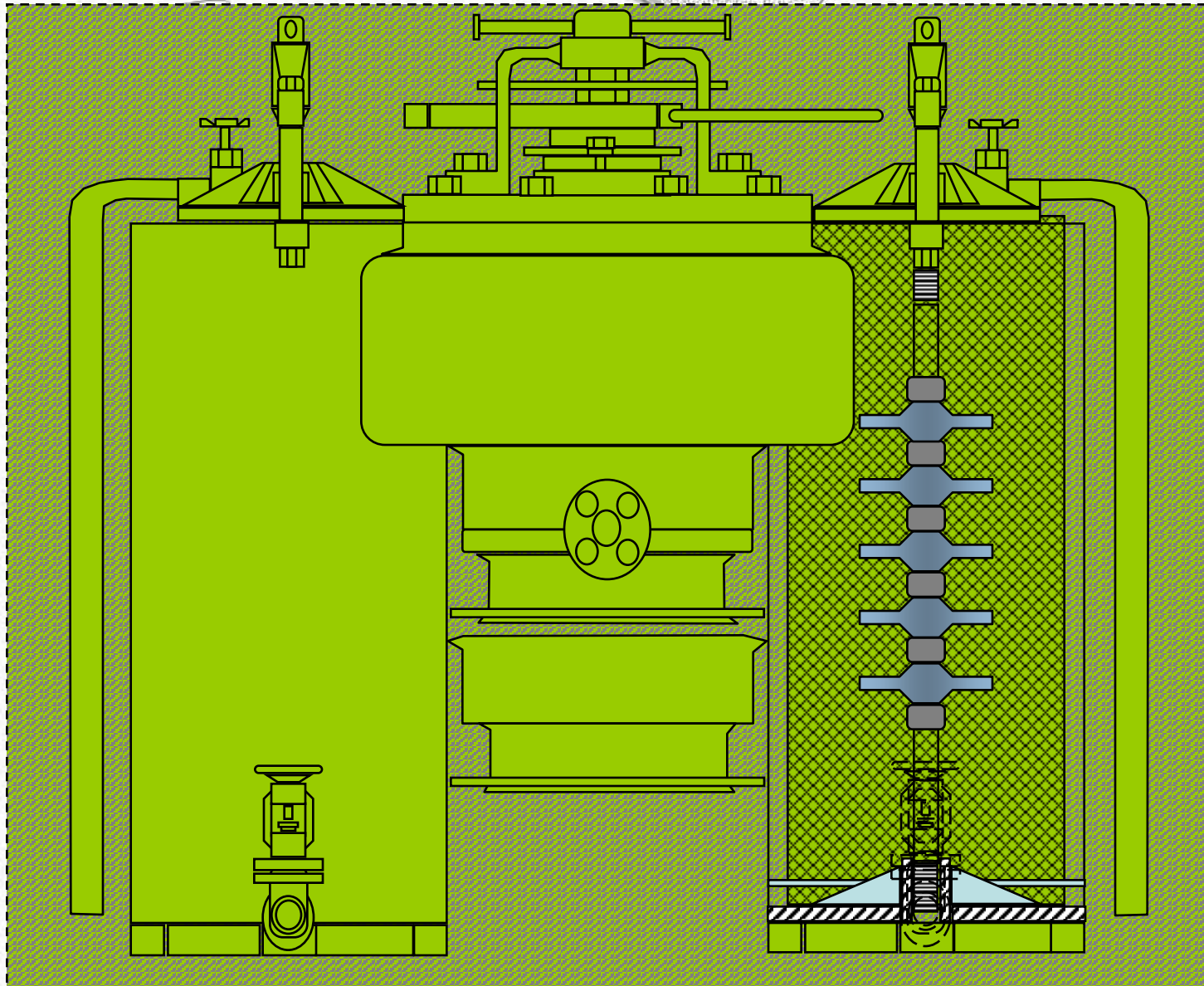




SYSTEM COMPONENTS AND COMPONENT PARTS (cont'd)

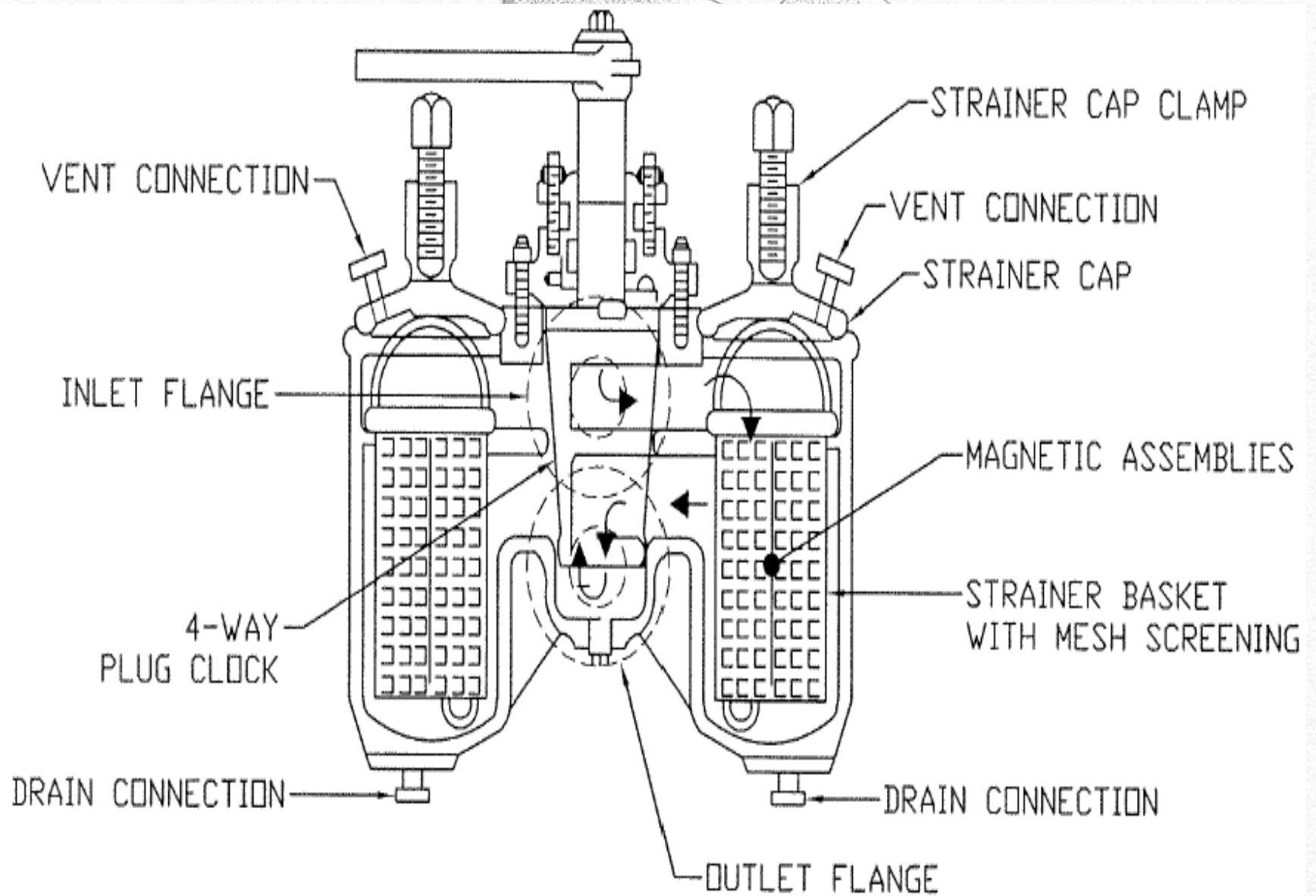
- ***Lube oil strainer/cover-*** The duplex strainer is located downstream from the combined pump discharge header. The strainer basket is double walled, fitted with removable magnet assemblies that attract and hold ferrous particles, and has mesh screening that strains out nonferrous particles. The strainer body has one oil inlet and one oil outlet and is equipped with vent and drain connections.

DUPLEX STRAINERS



Lube oil strainer/cover (cont'd)

- A threaded cap clamp holds the strainer cap in place. An integral fourway plug cock valve switches the flow of lubricating oil from one basket to the other. One basket can be removed for cleaning without interruption of flow through the strainer. A differential pressure gauge is provided for indicating the inlet and outlet pressure differential. The lubricating oil strainers are encased in a sheet metal housing to deflect oil spray in the event of a leak.



How do we prevent contamination???

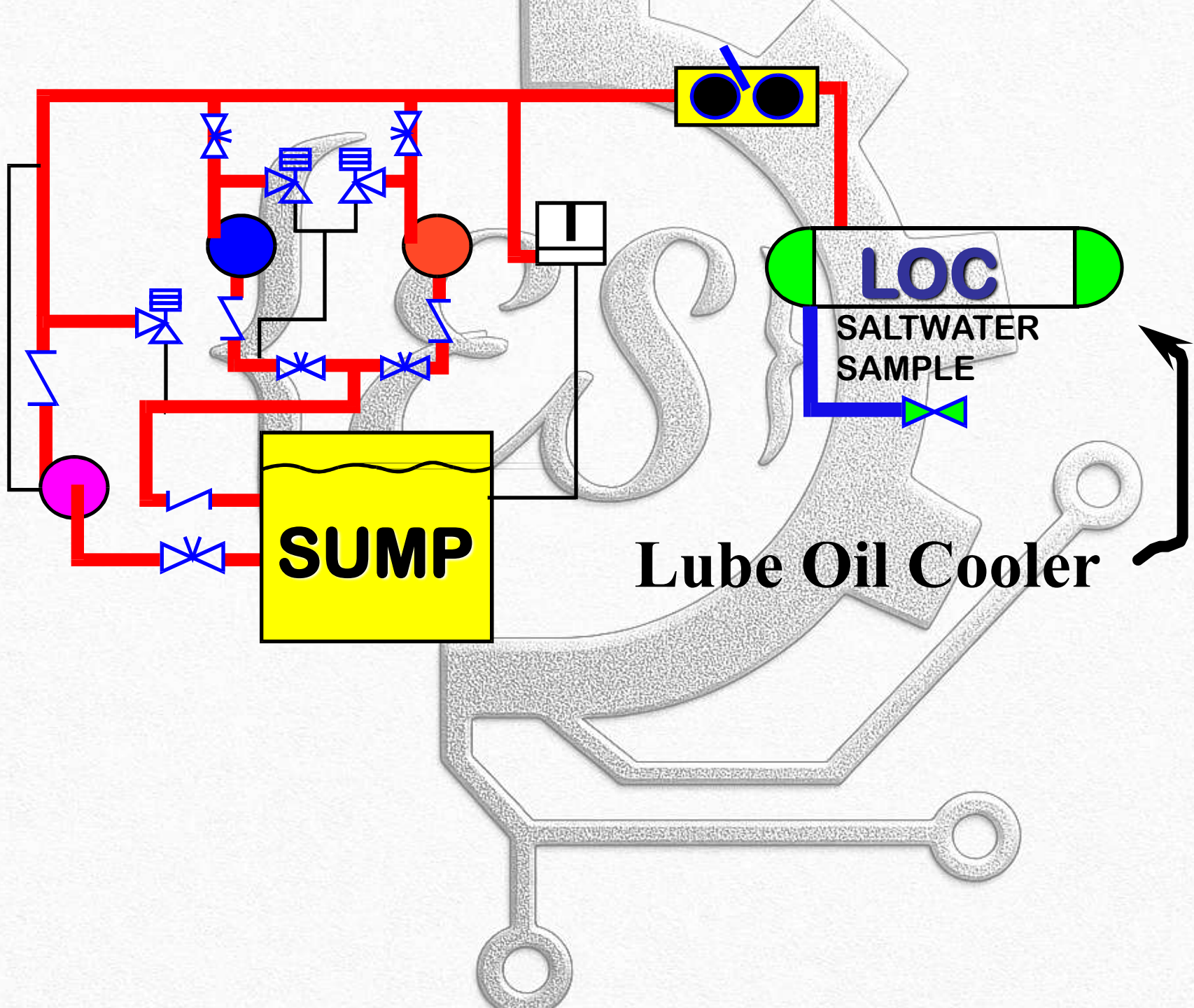
- Shift **I**nspect and **C**lean **L**ube **O**il **S**trainers (**SICLOS**).
- Sample and monitor
- Renovate in a Purifier
- DUMP

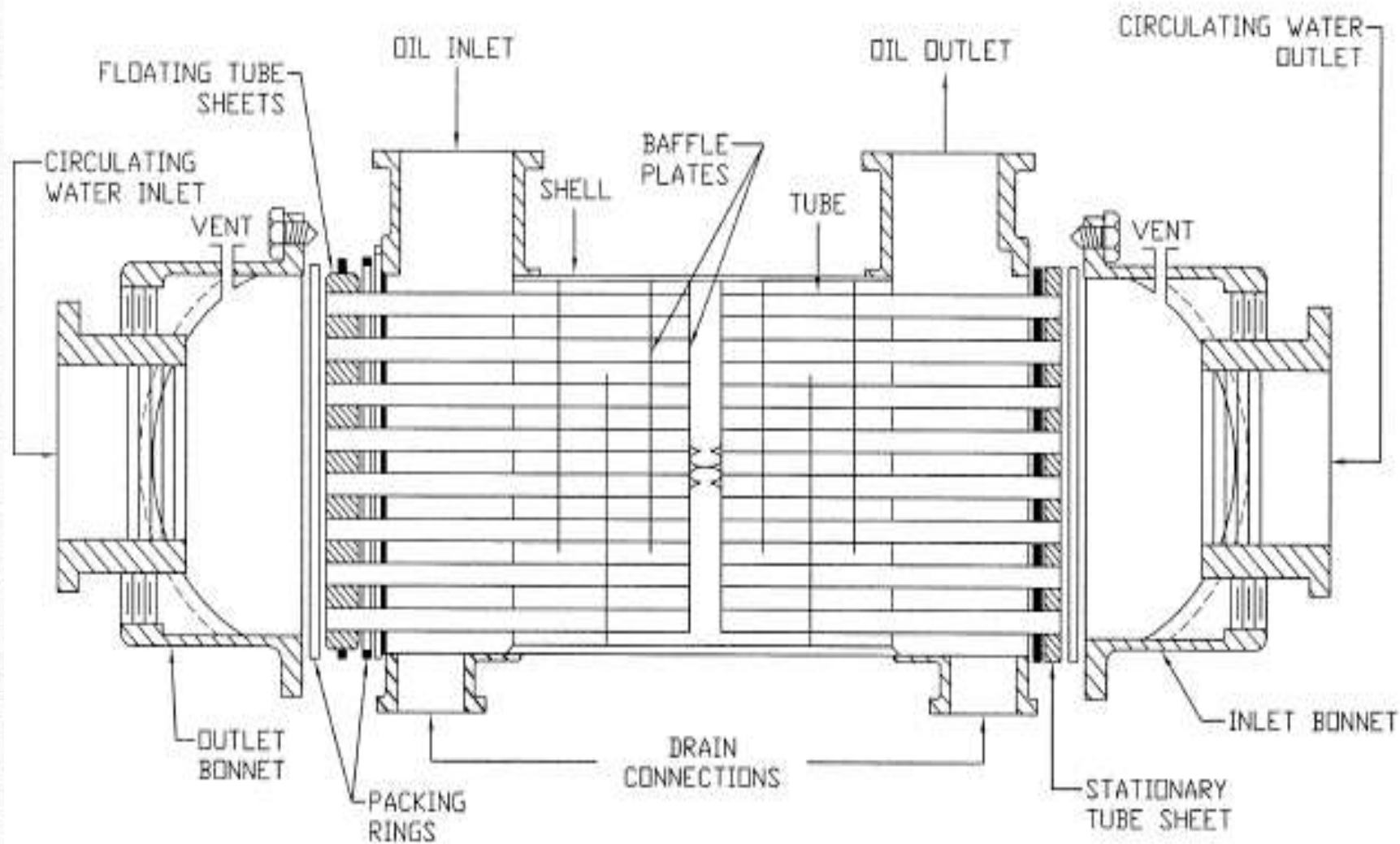
SICLOS CP# LODS

- Daily, while operating
- Every 4 hours when:
 - Pressure differential > normal
 - Abnormal system operation (i.e. high bearing temps)
 - for 24 hrs following shutdown > 1 wk
 - for 48 hrs following repairs to LO System components
- IAW EOSS

SICLOS

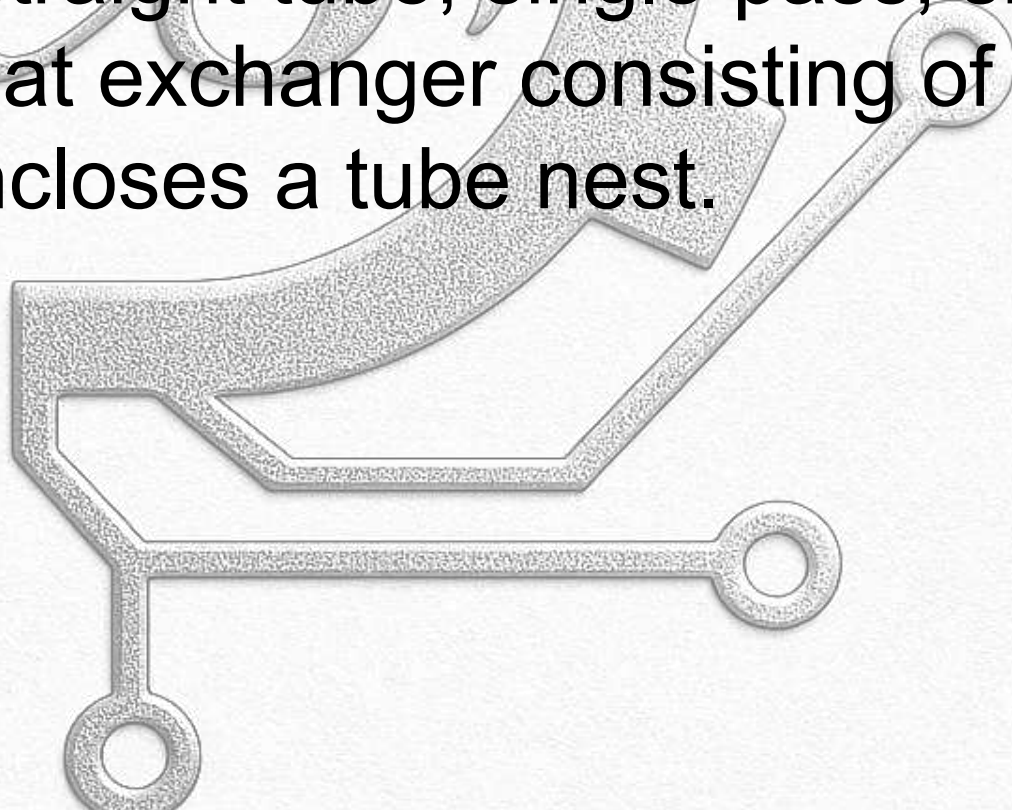
- ⊗ **FOR THE FIRST 24 HOURS IF THE SYSTEM HAS BEEN SHUT-DOWN > THAN 1 WEEK**
- ⊗ **FOR THE FIRST 48 HRS IF SYSTEM INTEGRITY HAS BEEN BROKEN.**
- ⊗ **WHILE IN HEAVY SEAS**





SYSTEM COMPONENTS AND COMPONENT PARTS (cont'd)

- **Main lube oil cooler-** The American Standard main lubricating oil cooler is a horizontal, straight tube, single-pass, shell-and-tube heat exchanger consisting of a shell that encloses a tube nest.



LUBE OIL OUT

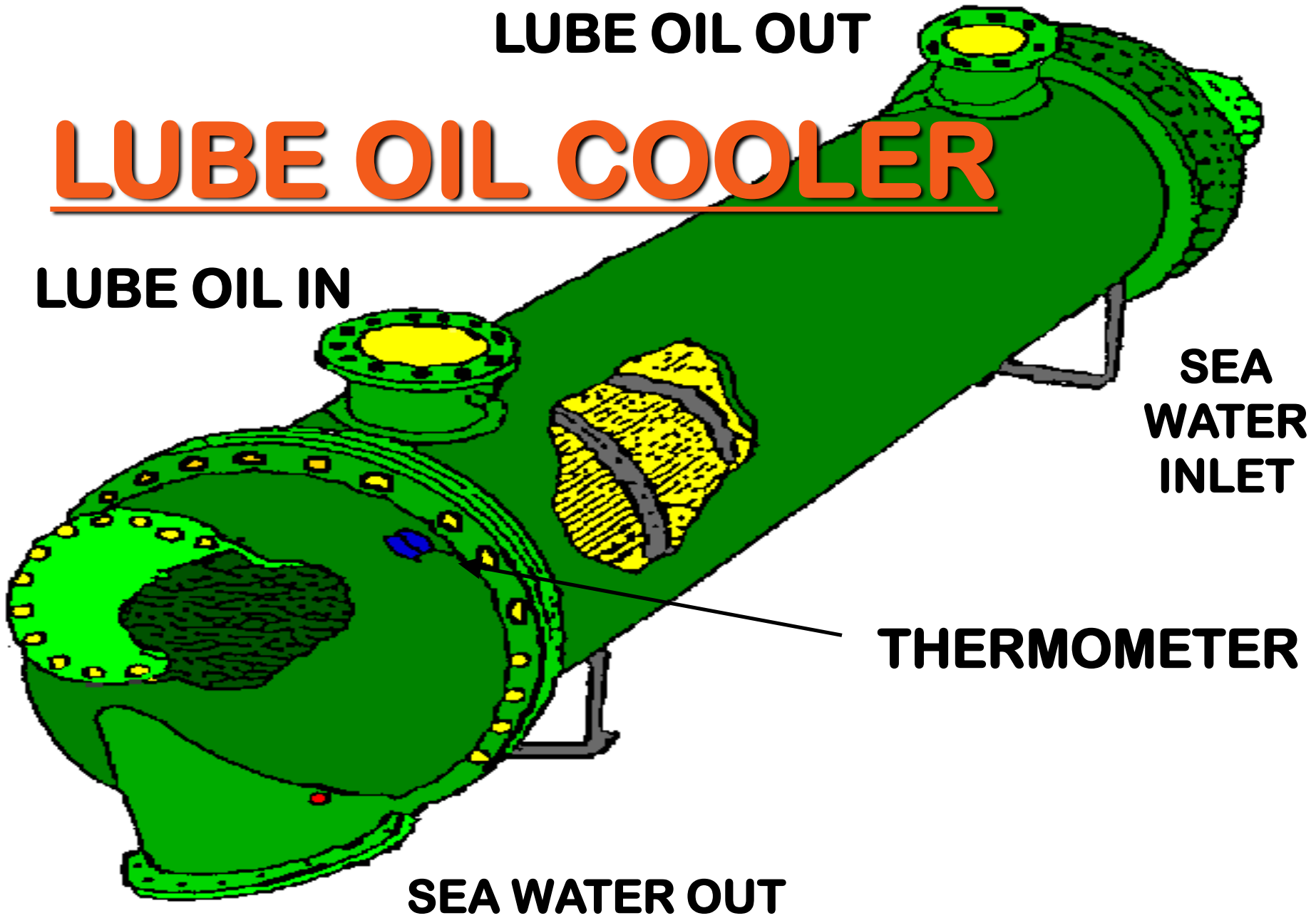
LUBE OIL COOLER

LUBE OIL IN

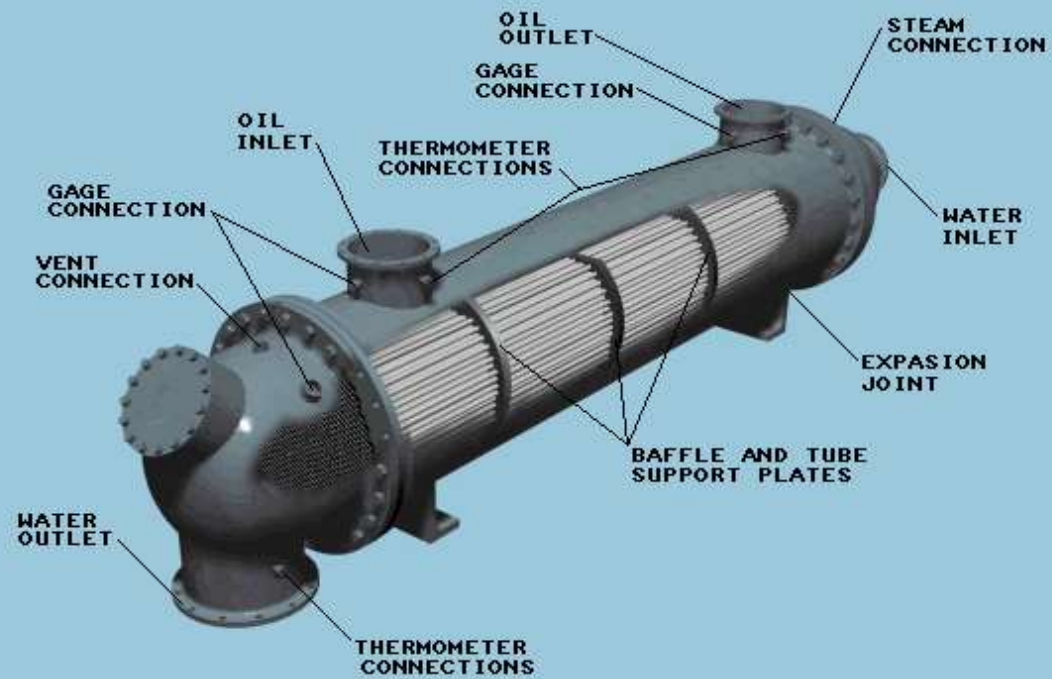
**SEA
WATER
INLET**

THERMOMETER

SEA WATER OUT

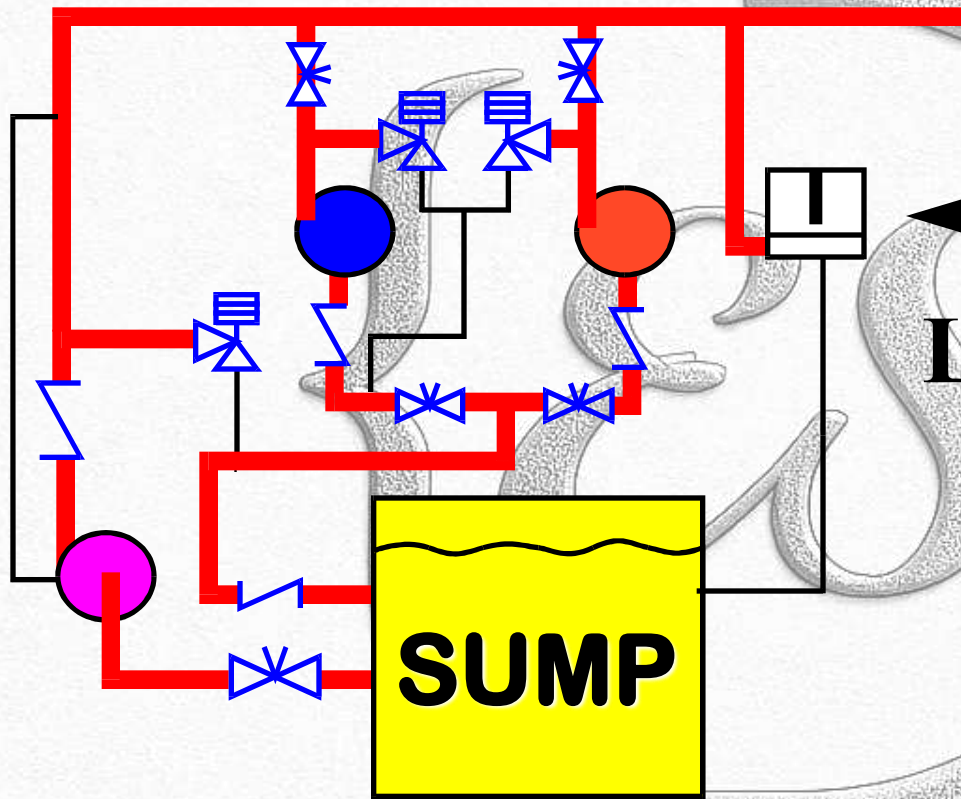


Main Lube Oil Cooler



SYSTEM COMPONENTS AND COMPONENT PARTS (cont'd)

- **Relief valve**— The attached and standby pumps discharge into 6-inch piping to the lubricating oil system piping. The standby pump has a 4-inch relief valve in the line. The motor-driven pump has a 4-inch discharge line leading to the lubricating oil system. Relief valves are in the discharge line ahead of the pump discharge valves. Each relief valve is set for 65 psig and relieves excess oil to the sump.

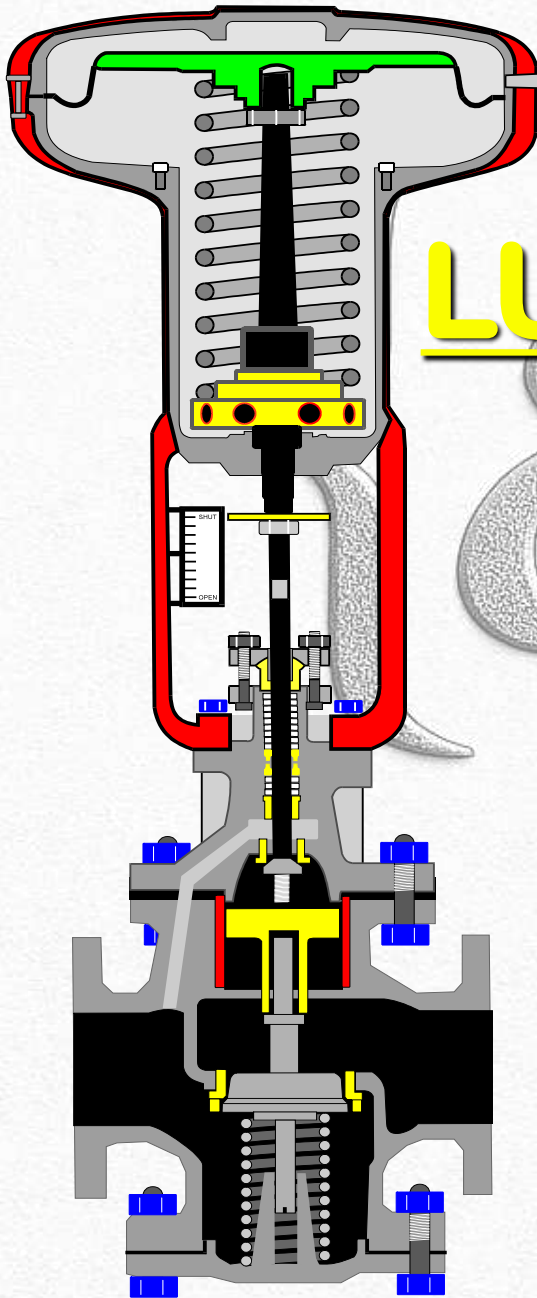


Lube Oil Unloader

- **Maintains 15 psig at Most Remote Bearing**

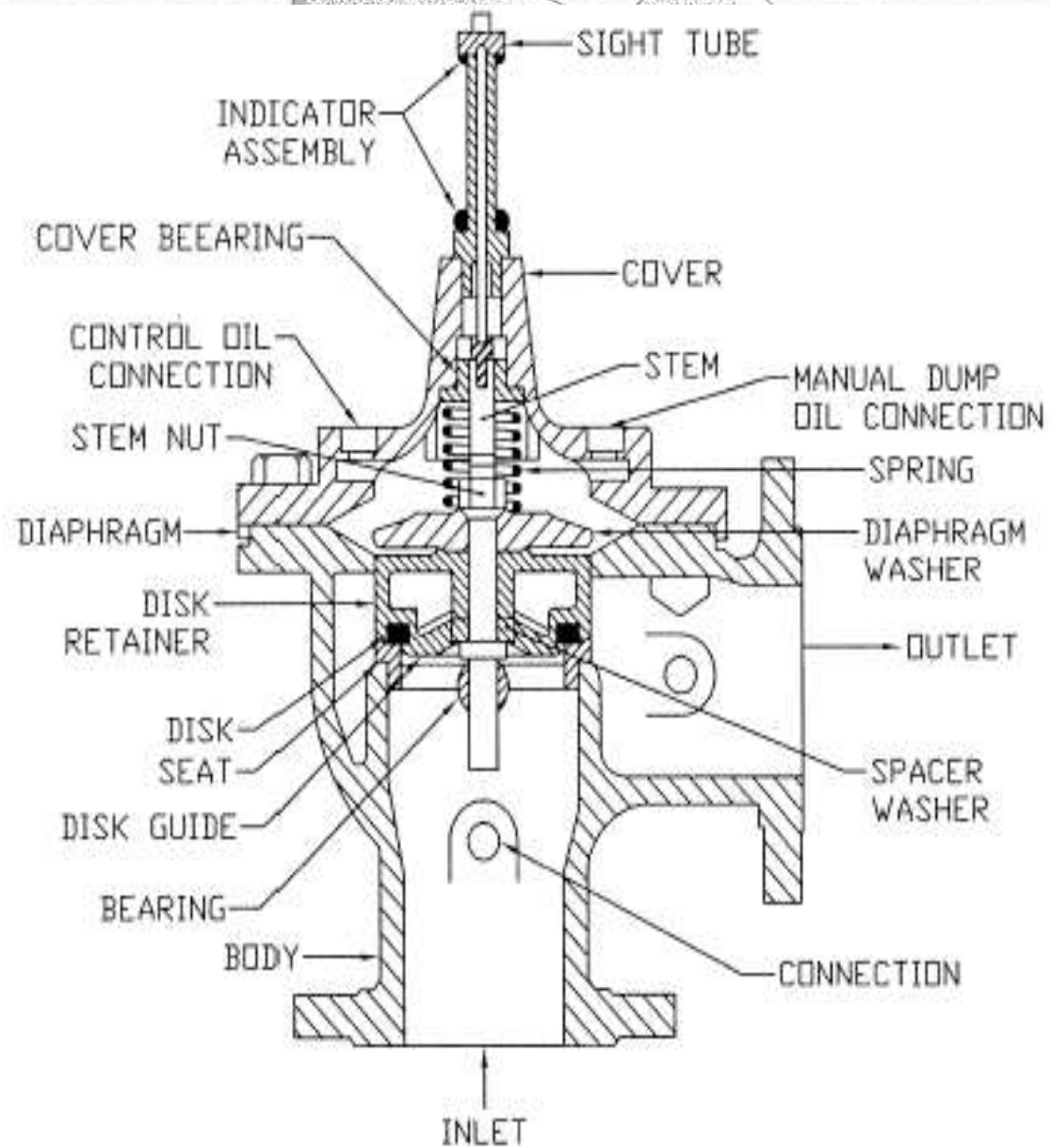
SYSTEM COMPONENTS AND COMPONENT PARTS (cont'd)

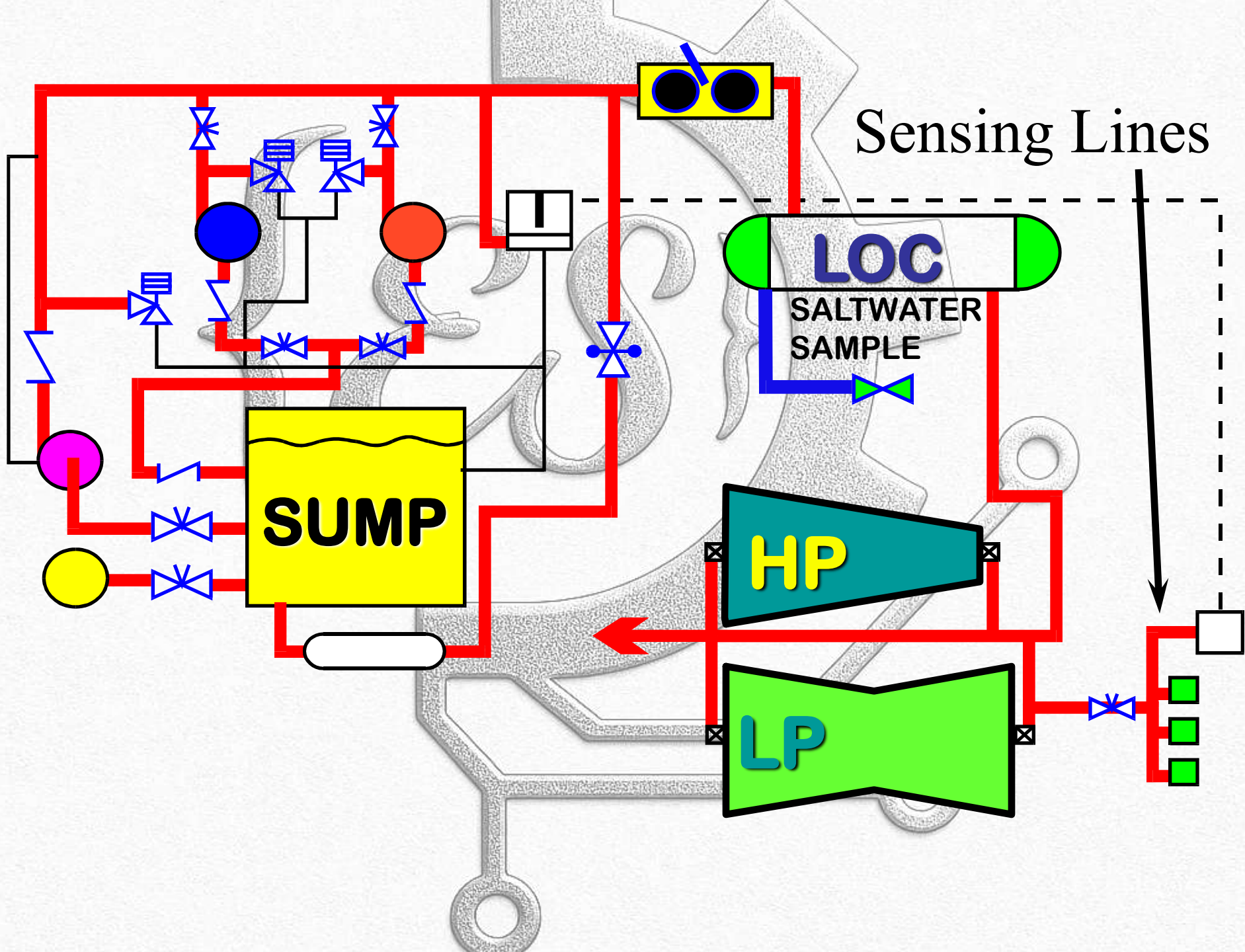
- ***Unloading valve-*** The 4-inch unloading valve and its associated pilot valve are installed in the combination discharge main from the three main lubricating oil service pumps. The 4-inch unloading valve is an angle stop, hydraulically operated, pilot controlled, diaphragm-actuated assembly of modified globe design.



LUBE OIL UNLOADER

- Leslie type
- Air or Mechanically actuated
- Initiated by
Most Remote Bearing pressure
- Return oil to sump





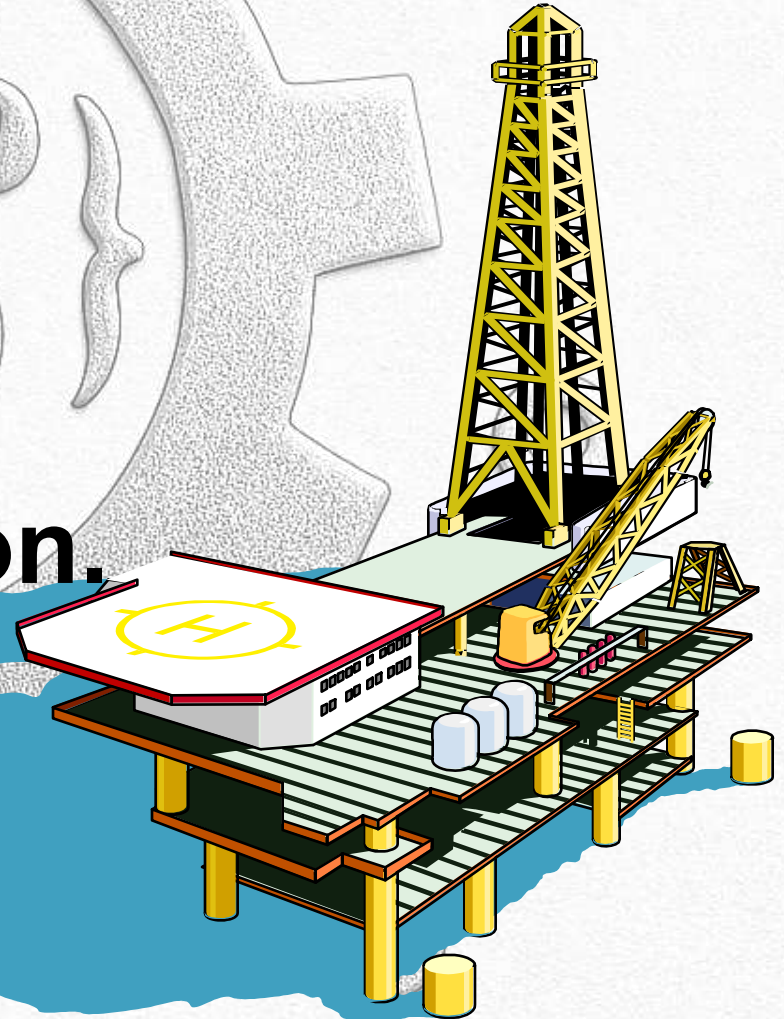


SENSING LINES

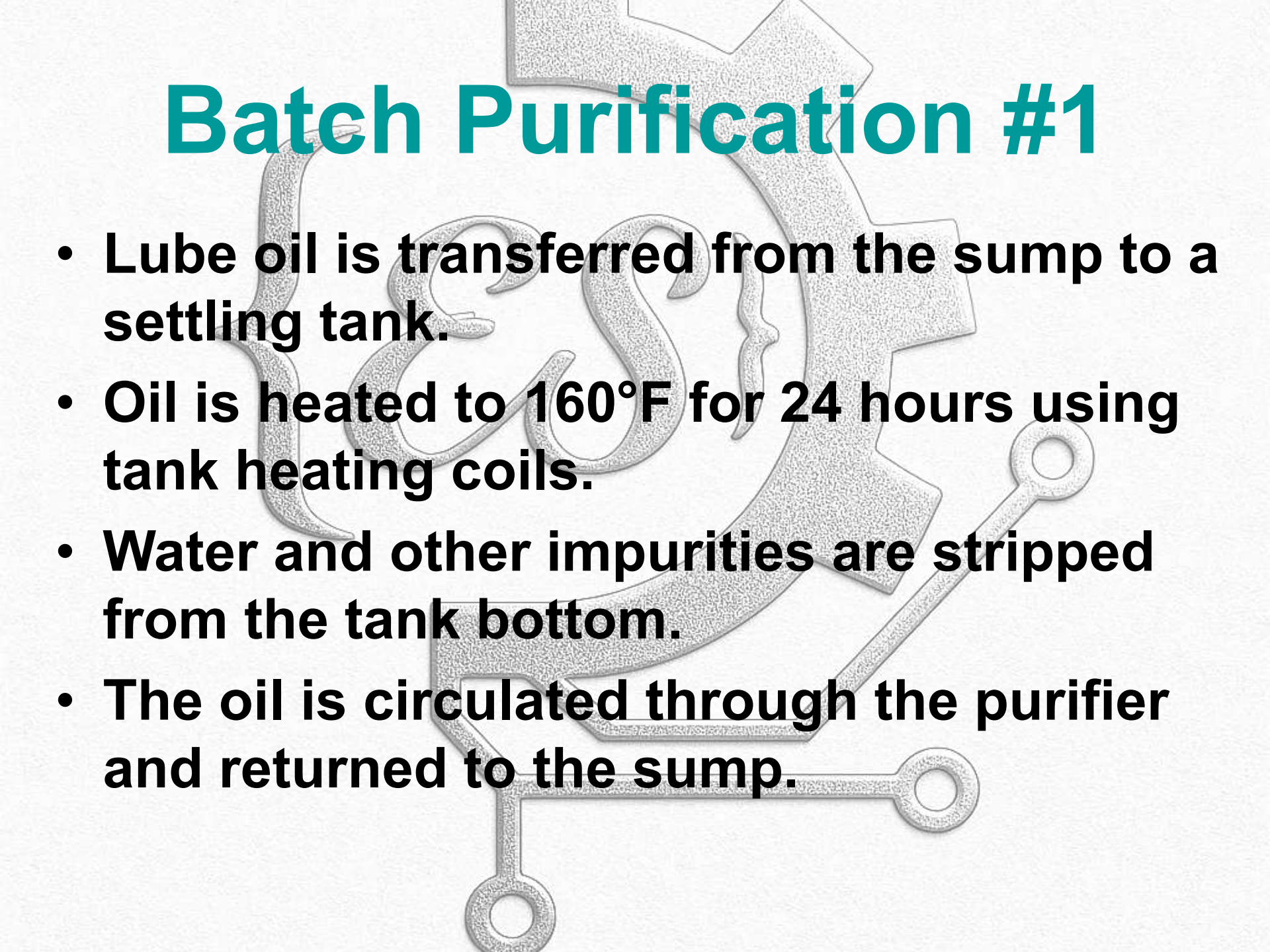
- PRESSURE SWITCHES FOR ELOP'S
- PRESSURE SWITCH FOR L/A ARM
- SENSING LINE TO L/O UNLOADER
- LUBE OIL PRESSURE GAGES

Purification Methods

- Two purification methods:
 - **Batch** purification.
 - **Continuous** purification.



Batch Purification #1

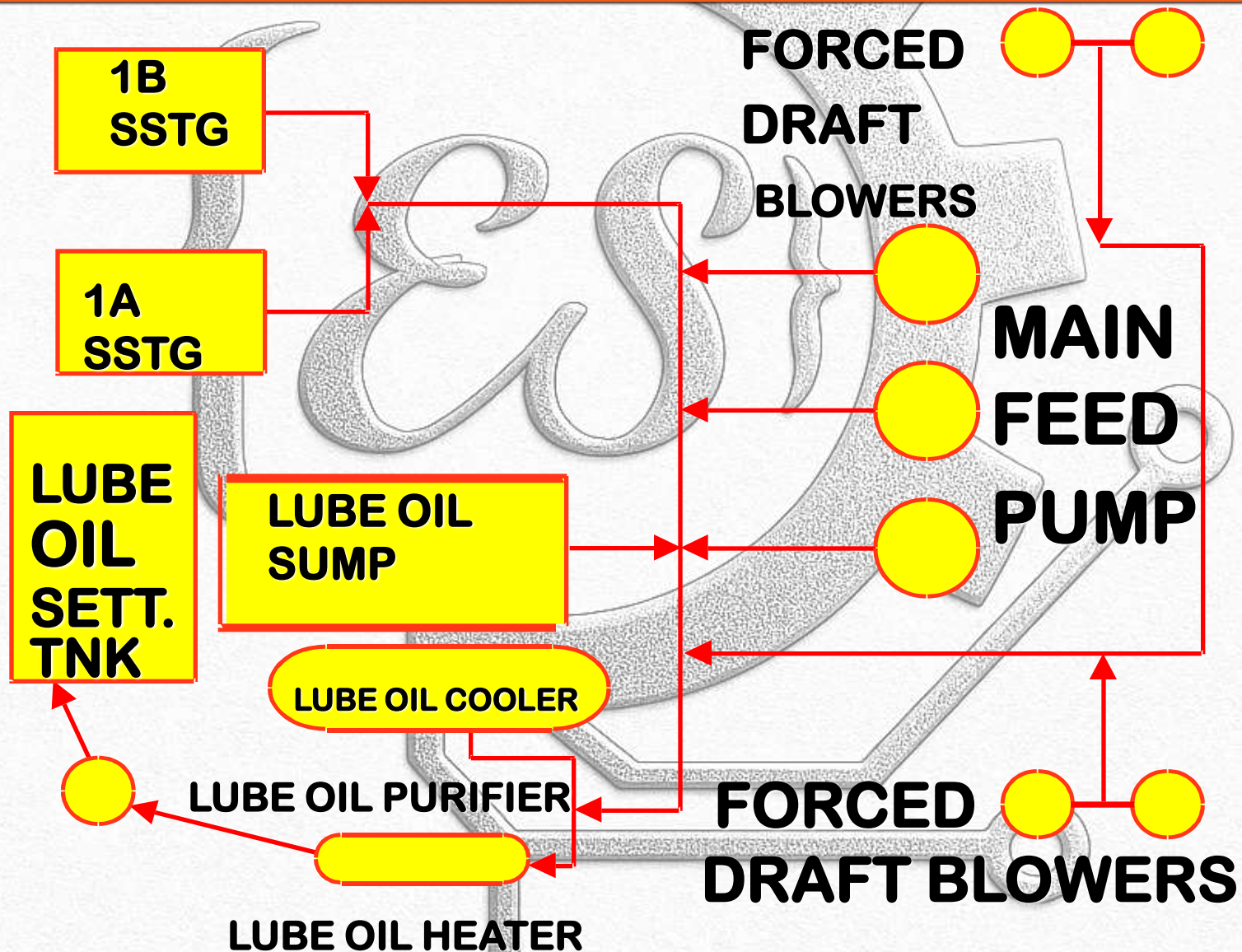


- **Lube oil is transferred from the sump to a settling tank.**
- **Oil is heated to 160°F for 24 hours using tank heating coils.**
- **Water and other impurities are stripped from the tank bottom.**
- **The oil is circulated through the purifier and returned to the sump.**

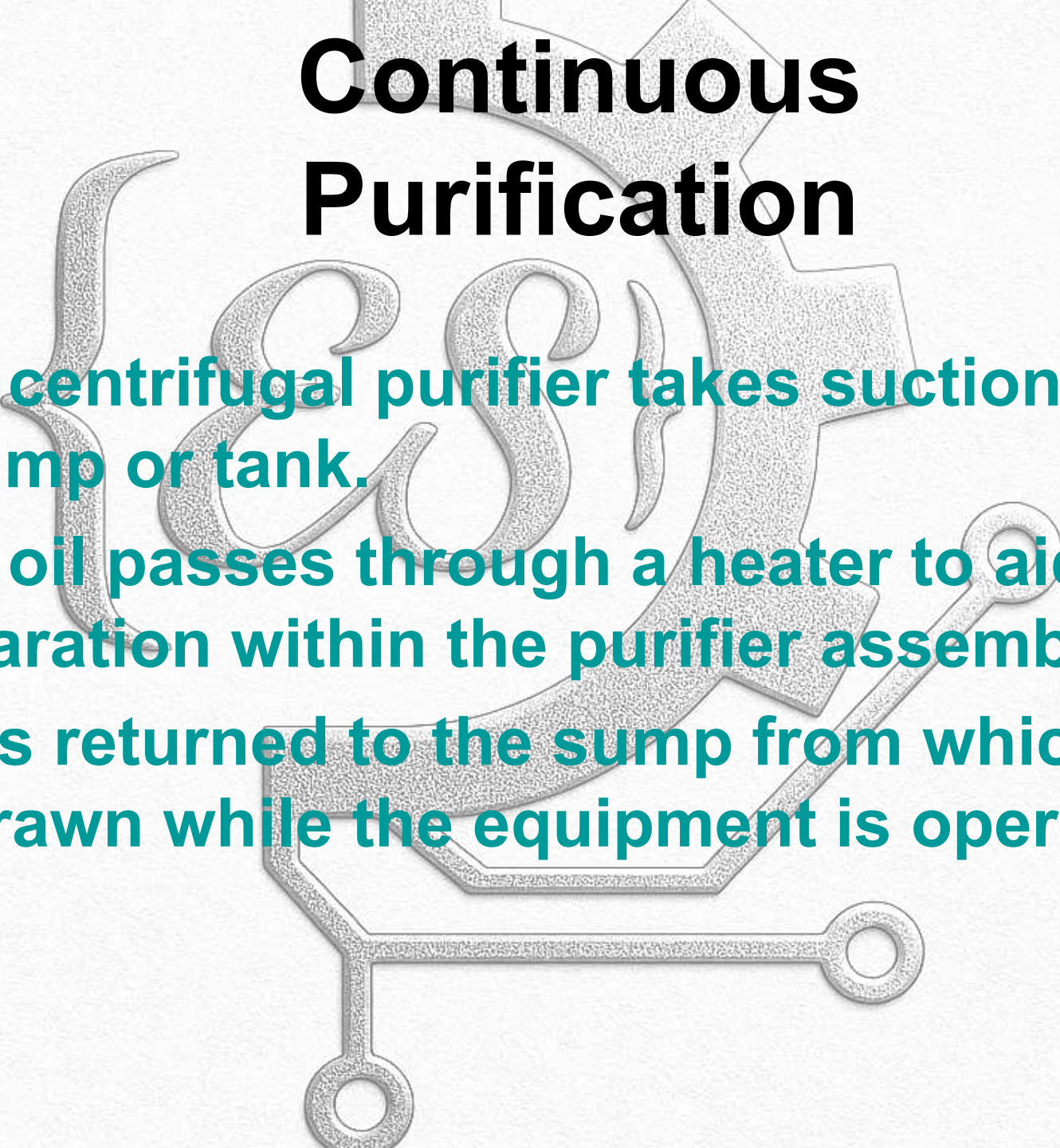
Batch Purification #2

- **Allows the equipment to be placed back into operation without having to wait 24 hours.**
- **Instead of leaving the sump empty while renovating, new oil is struck down from a storage tank**
- **New oil should still be purified as it is transferred from tank to sump**

LUBE OIL PURIFICATION AND TRANSFER



Continuous Purification

A decorative background graphic featuring a large, stylized gear with a circuit-like line running through it. The gear is light gray and has a textured, metallic appearance. The circuit line is also light gray and has a similar texture. The overall design is industrial and technical.

- The centrifugal purifier takes suction from a sump or tank.
- The oil passes through a heater to aid in separation within the purifier assembly.
- Oil is returned to the sump from which it is drawn while the equipment is operating.

METHODS OF PURIFICATION

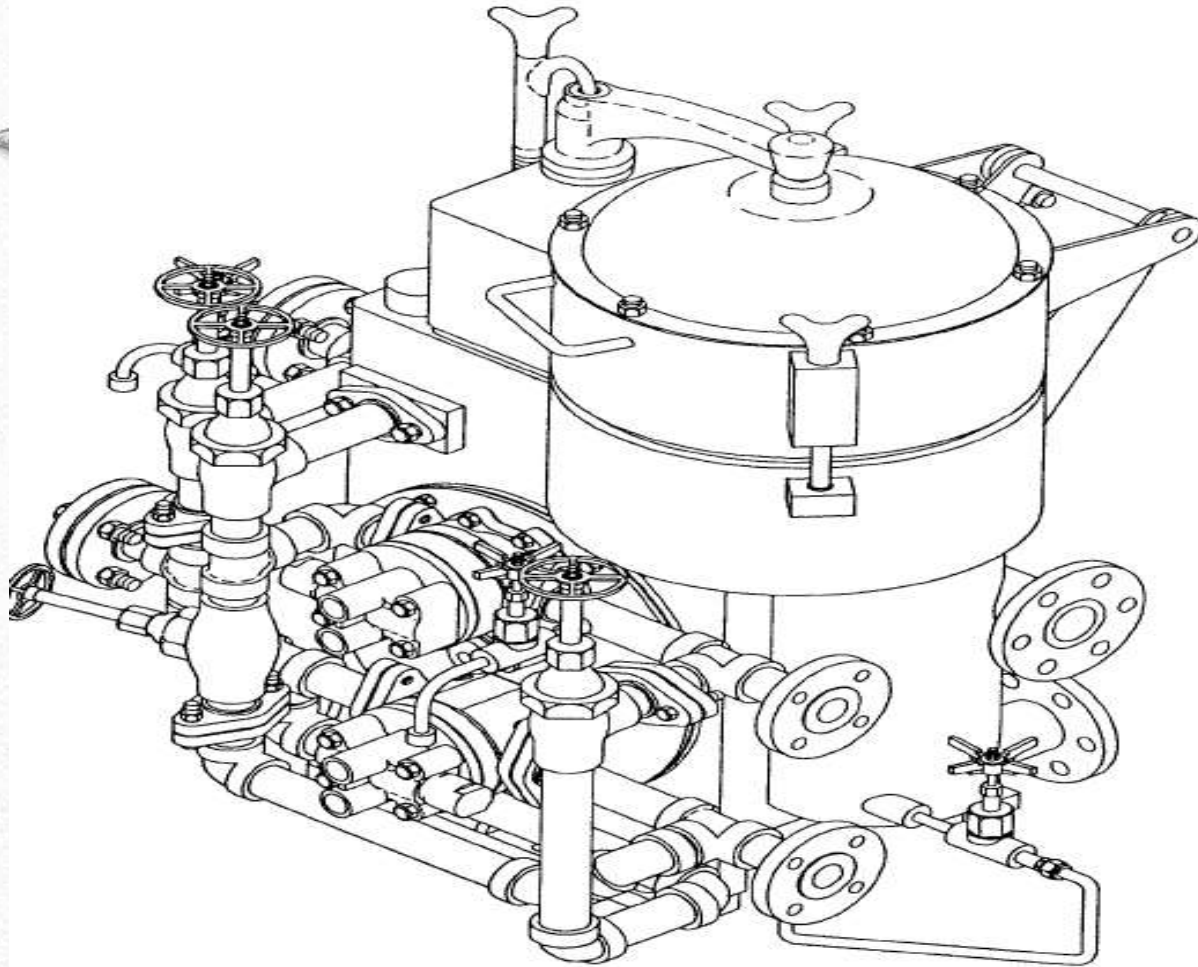
CONTINUOUS

- **MAIN SUMP ONLY.**
- **THE PURIFIER IS USED TO REMOVE IMPURITIES FROM THE LUBE OIL.**

BATCH

- **LUBE OIL SETTLING TANK.**
- **HEAT TO 160°F AND LET IT SETTLE FOR 24 HOURS.**

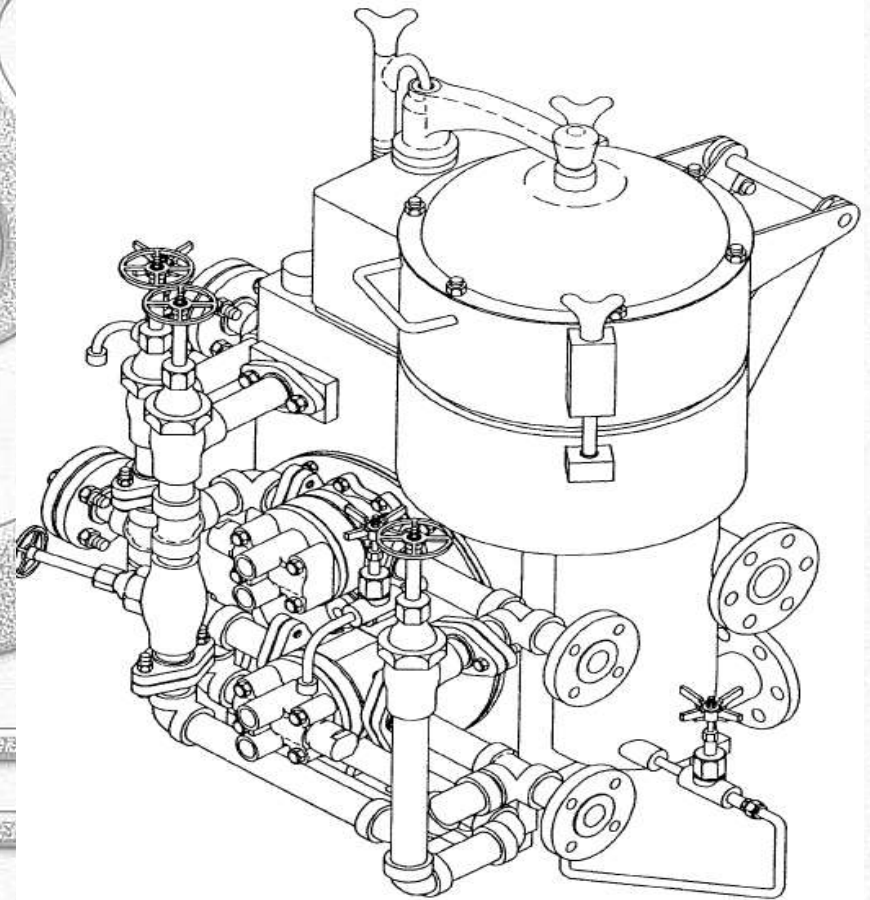
THE LUBE OIL PURIFIER



SYSTEM COMPONENTS AND COMPONENT PARTS (cont'd)

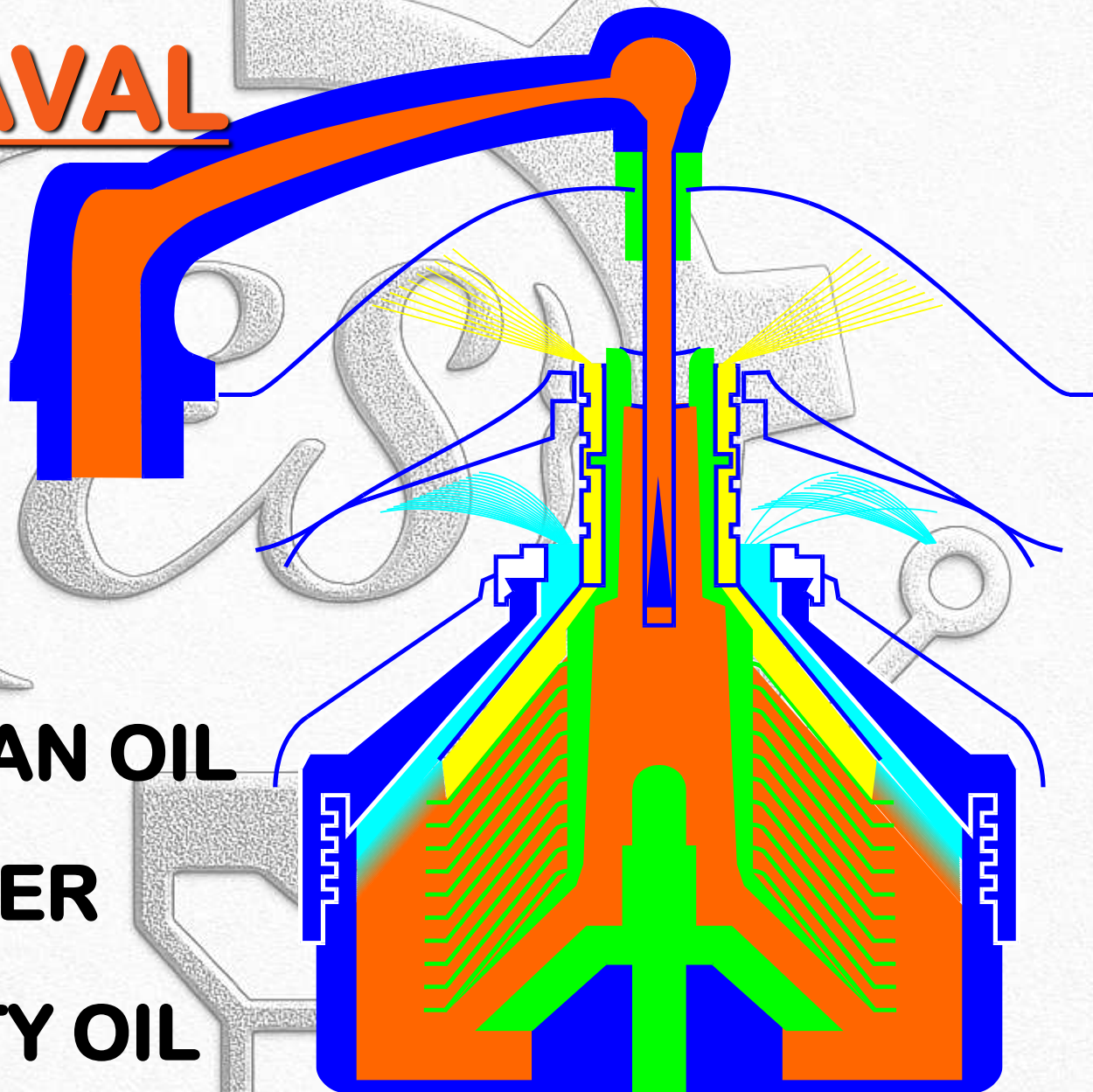
- ***Lube oil purifier-***

The Alpha-Laval lubricating oil purifier applies centrifugal force to oil to separate it into water, solids, and pure oil.



DELAVAL

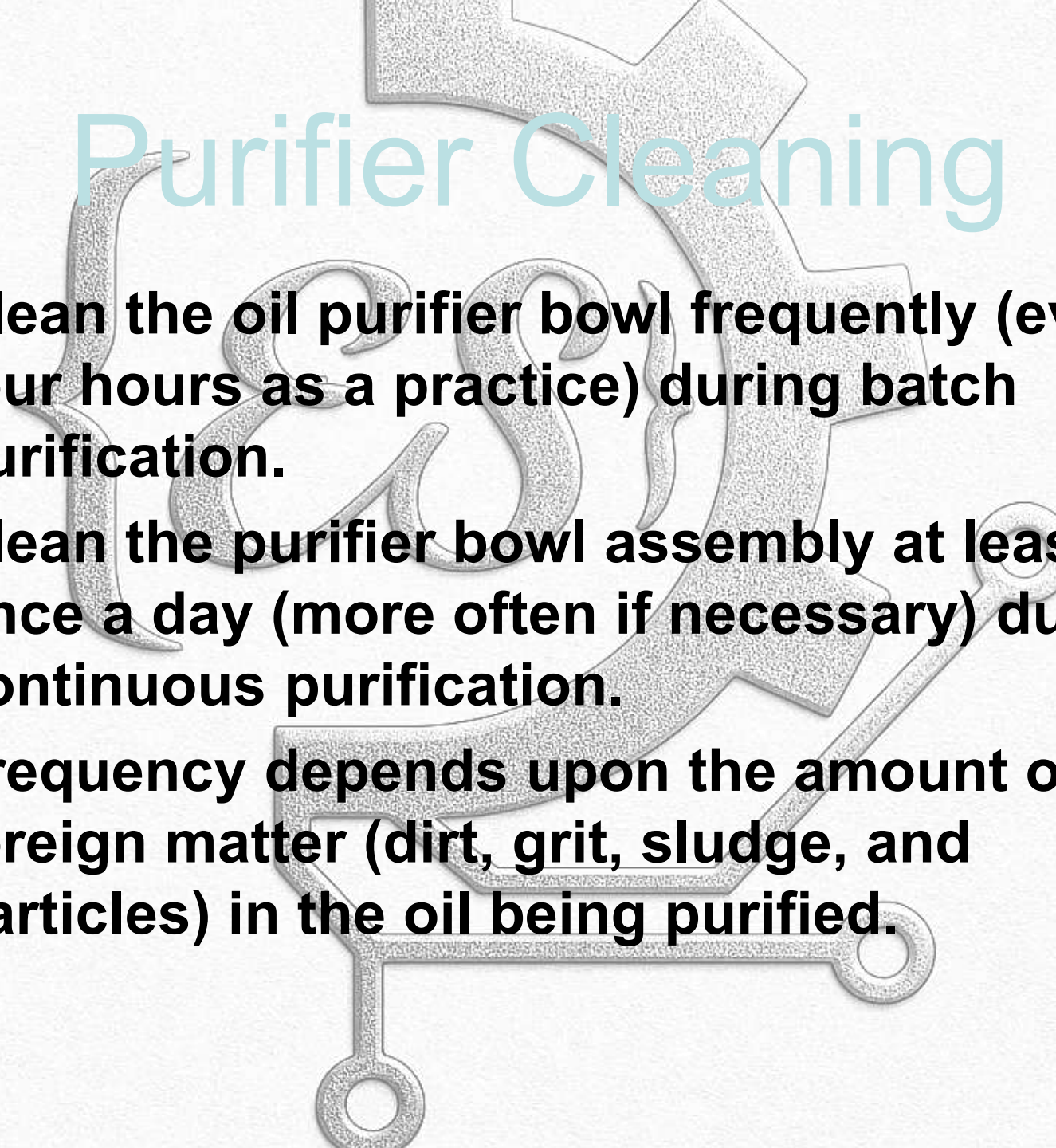
-  **CLEAN OIL**
-  **WATER**
-  **DIRTY OIL**



Methods of Operation:

- Separator: The centrifuge is adjusted to separate and discharge two liquids.
- Clarifier: The centrifuge is adjusted to discharge a single liquid from which the solids have been removed.

Purifier Cleaning



- **Clean the oil purifier bowl frequently (every four hours as a practice) during batch purification.**
- **Clean the purifier bowl assembly at least once a day (more often if necessary) during continuous purification.**
- **Frequency depends upon the amount of foreign matter (dirt, grit, sludge, and particles) in the oil being purified.**

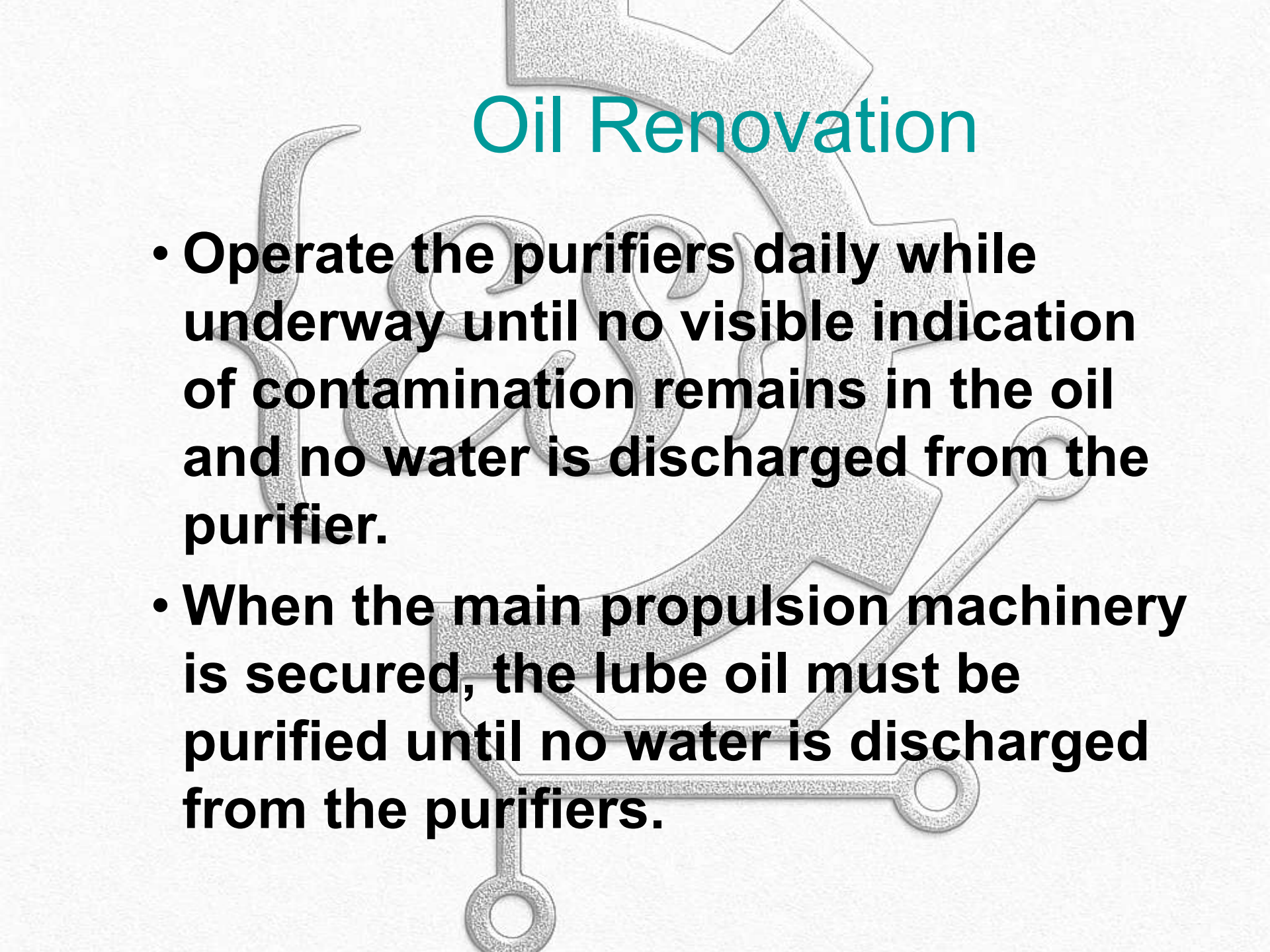
Purifier Sampling

- Draw oil samples from the inlet and outlet of the purifier for efficiency.
- The purified oil should be cleaner than the oil entering the purifier.(No Brainer)
- Water dischargeports should also be checked for oil.

Sample Frequency

- When oil is being renovated, efficiency samples should be drawn every 30 minutes.**
- When underway, with continuous purification in progress, sample once every 4 hours.**

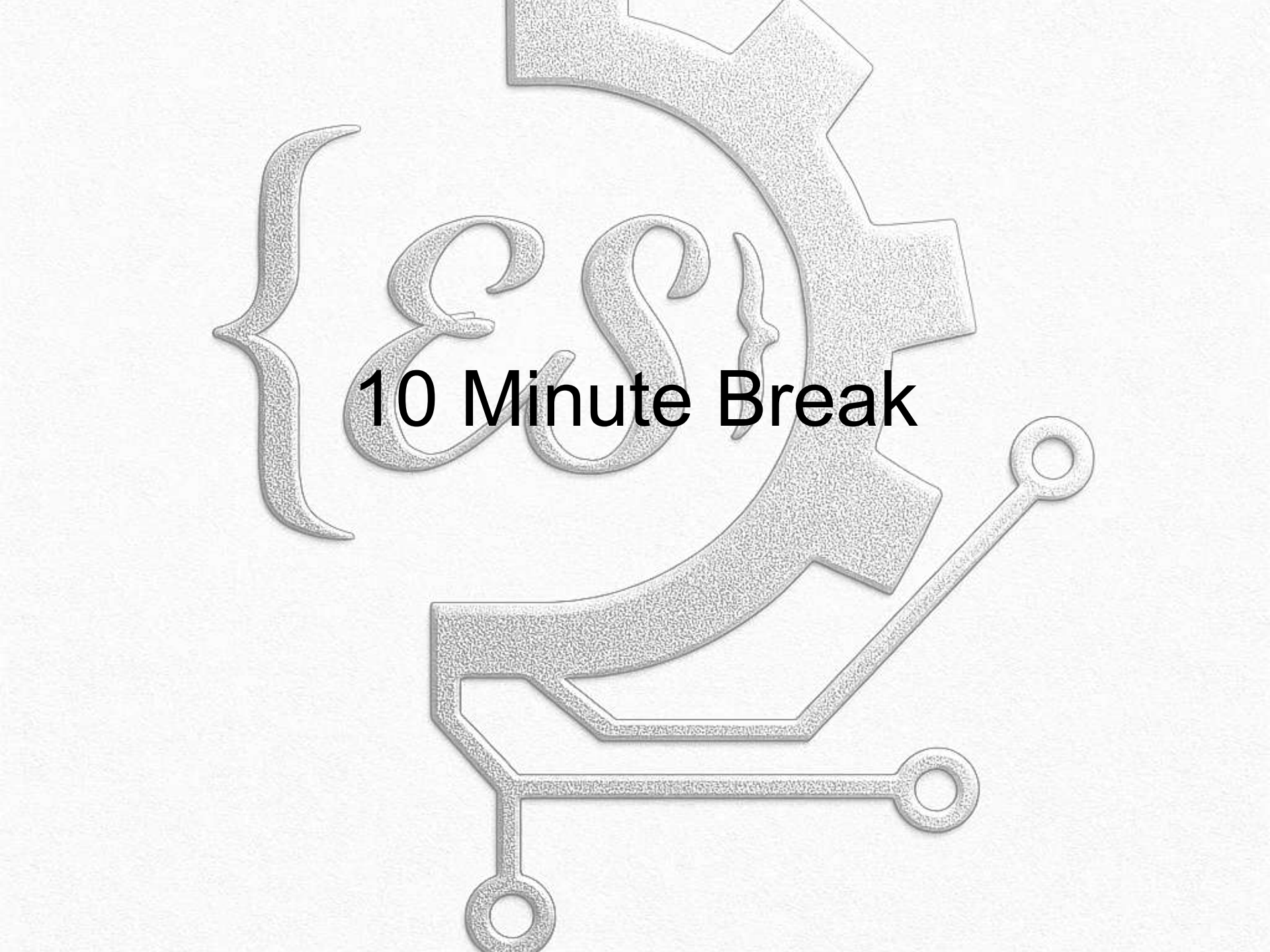
Oil Renovation

A large, faint, light gray graphic of a gear and a wrench is positioned in the background. The gear is centered and occupies most of the frame, with its teeth pointing towards the corners. A wrench is placed diagonally across the gear, with its head at the bottom left and its handle extending towards the top right.

- **Operate the purifiers daily while underway until no visible indication of contamination remains in the oil and no water is discharged from the purifier.**
- **When the main propulsion machinery is secured, the lube oil must be purified until no water is discharged from the purifiers.**

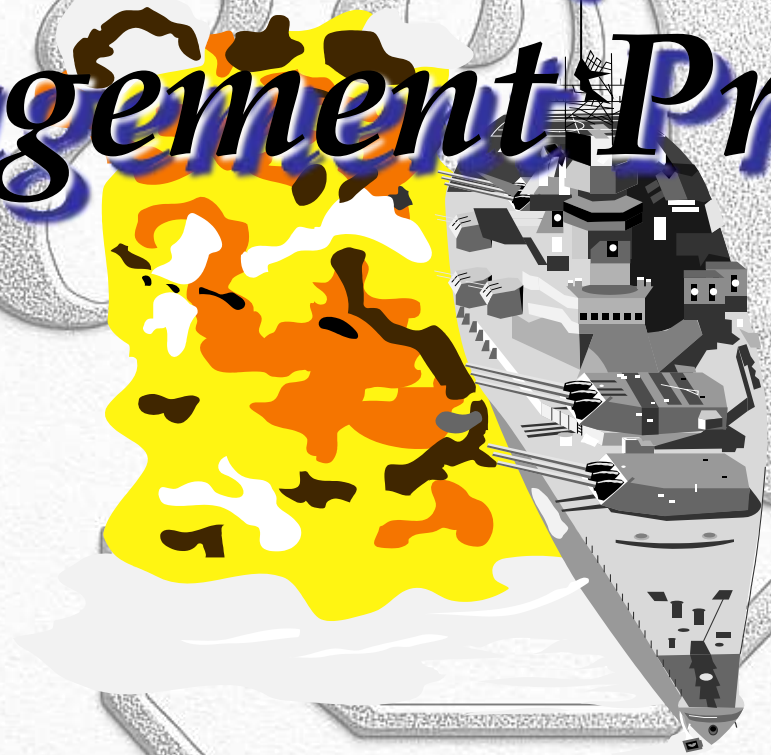
Formation Of Oil Emulsion:

- Gradual increase in acidity affected by operating temperature of the oil.
- The higher the operating temperature, the faster the breakdown of the oil.
- Emulsified oil does not form a proper oil film.
- Removal of water and other impurities from emulsified oil is difficult.



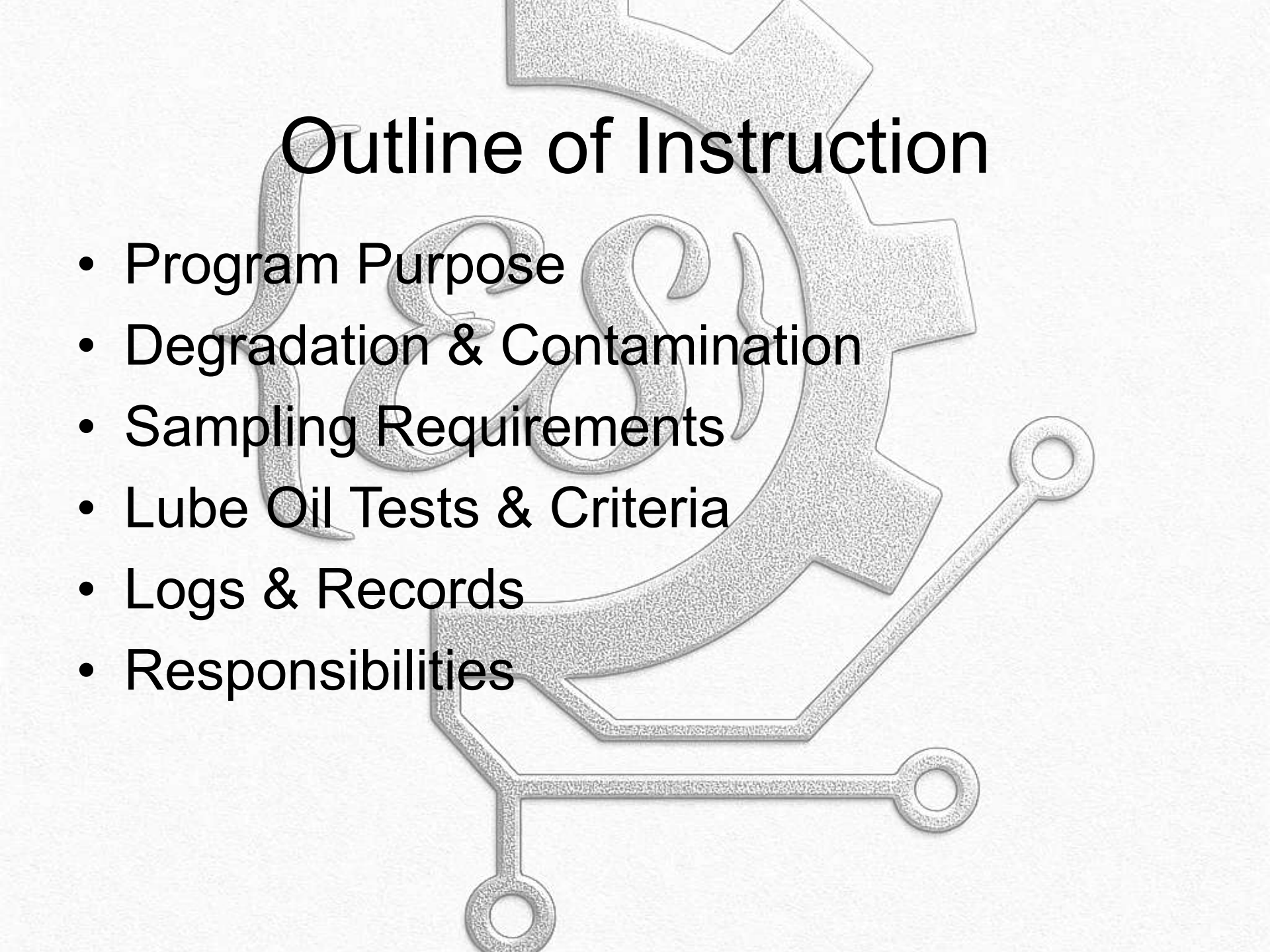
10 Minute Break

Lube Oil Quality Management Program



62A-103

Outline of Instruction

The background of the slide features a large, light gray gear with a textured, metallic appearance. Overlaid on the gear is a circuit board design, also in light gray, with several circular components and connecting lines. The overall aesthetic is technical and industrial.

- Program Purpose
- Degradation & Contamination
- Sampling Requirements
- Lube Oil Tests & Criteria
- Logs & Records
- Responsibilities

The LOQM Program

- Assesses Machinery Condition by Monitoring LO Quality
- Standardizes Sampling & Analysis
- Reduces Maintenance Time & Cost
 - Prevents Damage From Contaminated Oil
 - Early Detection Minimizes Overhauls
 - Eliminates Unwarranted Oil Changes

Lubricants Are Used For:



- Friction Reduction
- Heat Dissipation
- Corrosion Prevention

CAUSES OF OIL DEGRADATION

1. WATER

2. SEDIMENT

3. FUEL DILUTION

4. HEAT BREAKDOWN



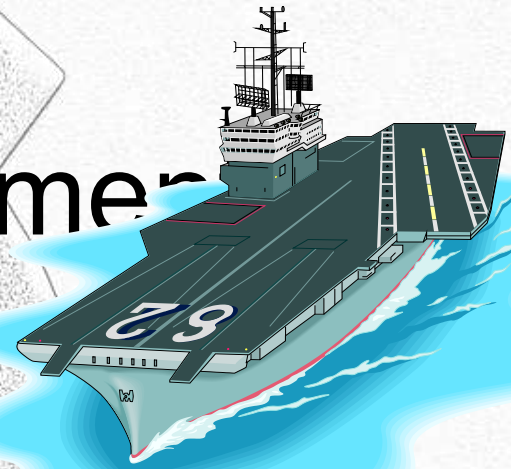
Water Contamination Sources

- CONDENSATION
- WORN STEAM SEALS
- IMPROPER START/STOP PROCEDURES
- SUMP CONTAMINATION



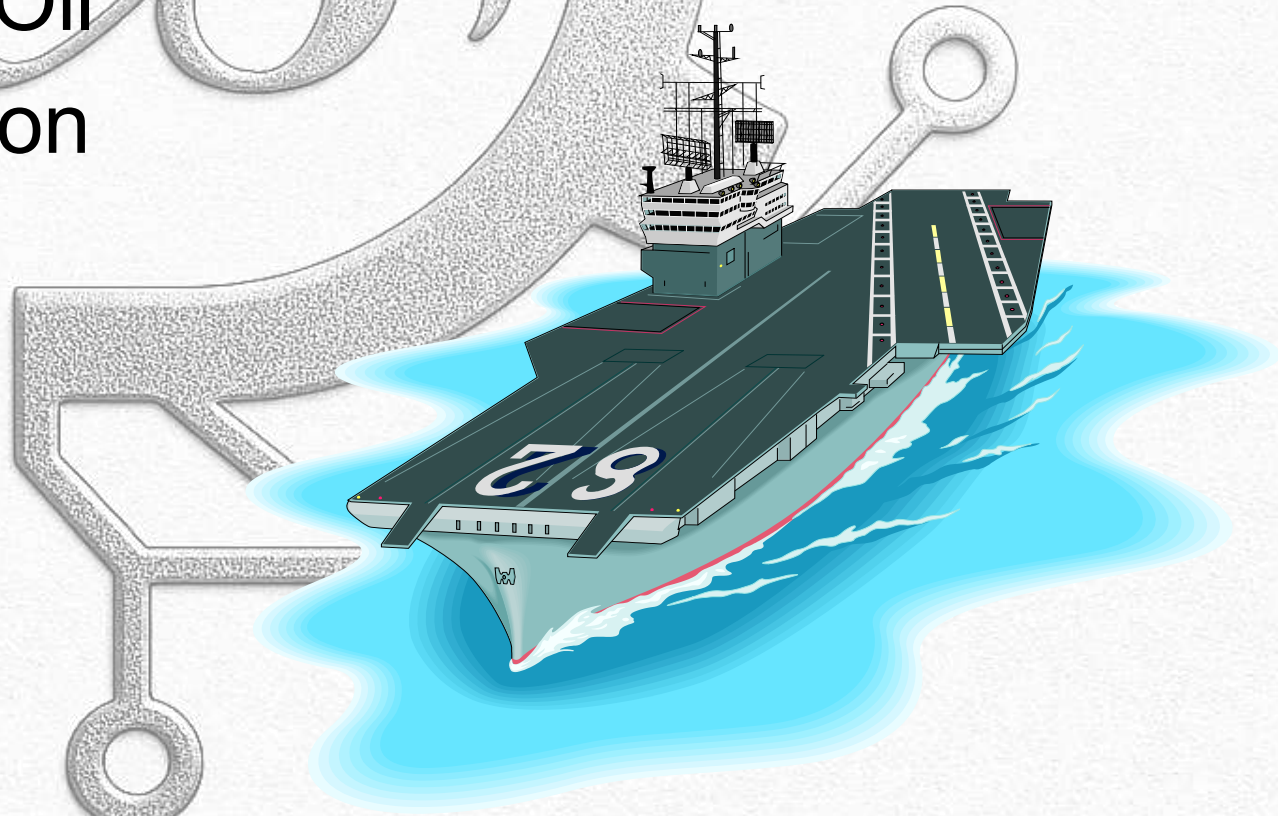
Sources of Sediment

- Enters through accesses during replenishment or inspection
- Result of wear and friction between system components.
- Causes pitting or scoring of surfaces.
- Affects viscosity and clogs filtering elements resulting in decreased oil flow and pressure.



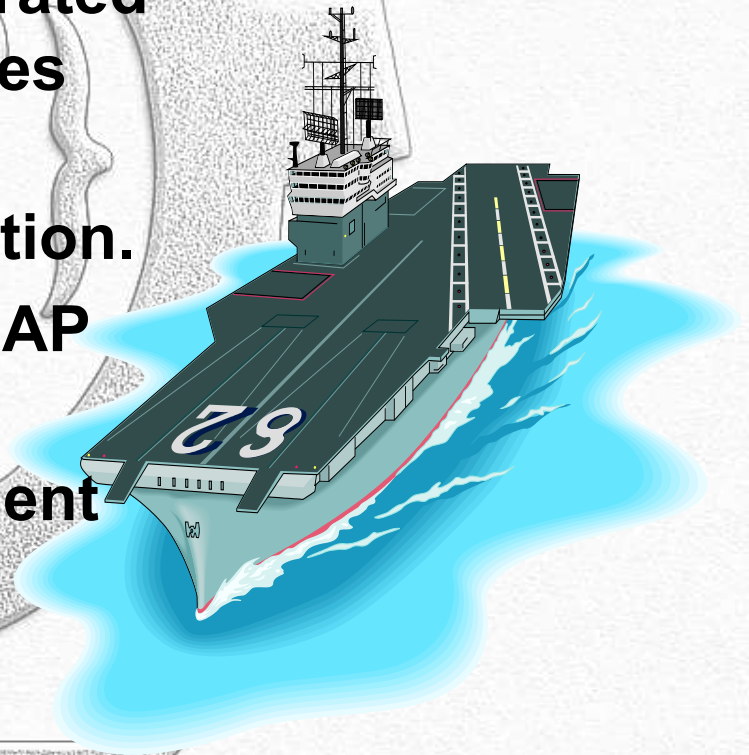
Fuel Dilution

- In Diesels
- Fuel blows into the Lubricating Oil
- Very Common

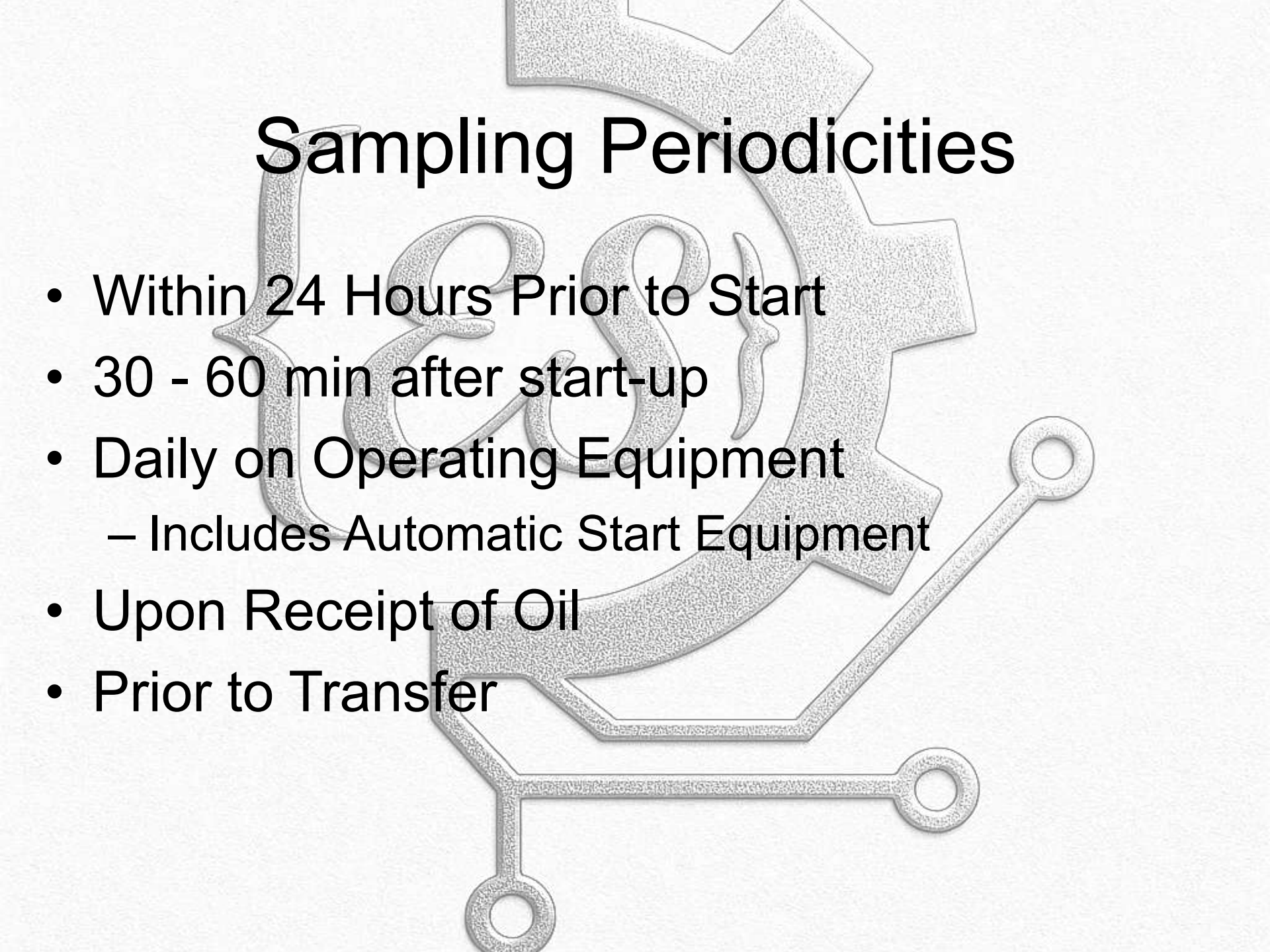


Thermal Breakdown

- **Caused by exceeding the rated temperatures and pressures for the oil.**
- **Accelerated by contamination.**
- **Should be detected by NOAP Sampling**
- **Usually requires replacement of the oil.**



Sampling Periodicities

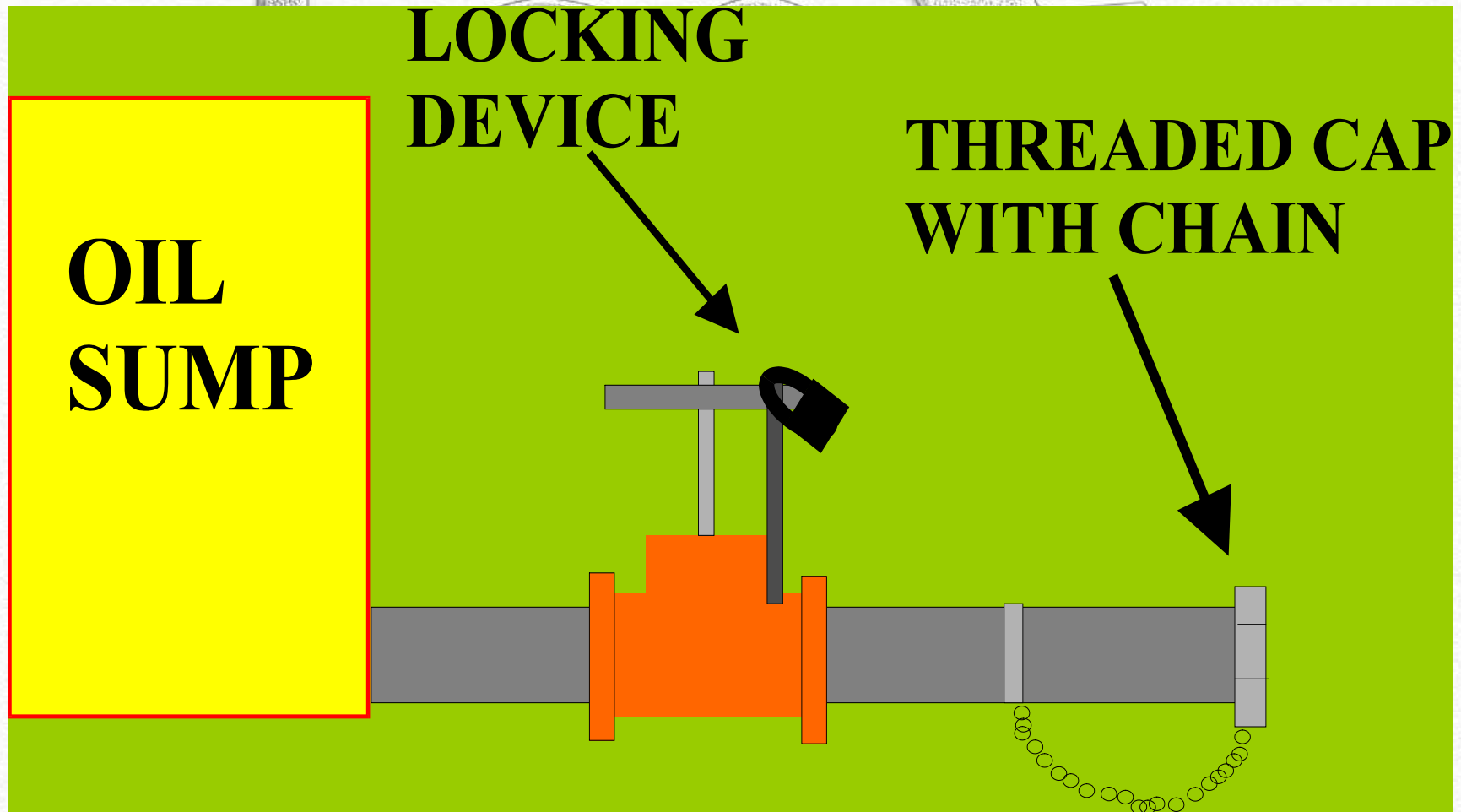
The background of the slide features a large, light gray gear with a circuit board pattern overlaid on it. The circuit board has several circular components and connecting lines, extending from the bottom and right sides of the gear.

- Within 24 Hours Prior to Start
- 30 - 60 min after start-up
- Daily on Operating Equipment
 - Includes Automatic Start Equipment
- Upon Receipt of Oil
- Prior to Transfer

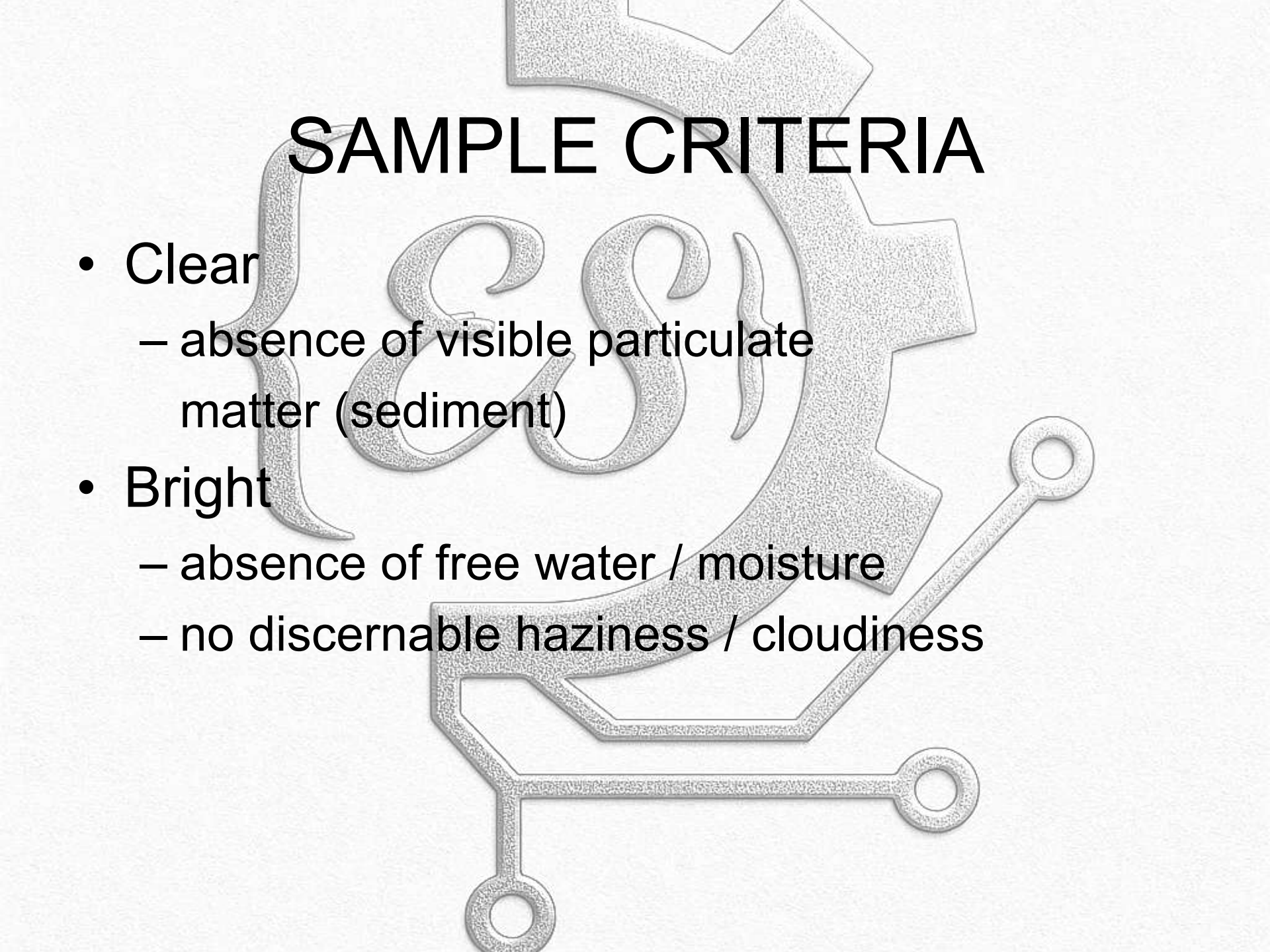
Sampling Periodicities (Cont.)

- Purifier Discharge 30 (10 - Rev 4) Minutes After Start Up and Every 4 Hours Thereafter
- Not Required on Equipment With Sumps Less Than 1 Gallon

Sample Connection



SAMPLE CRITERIA



- Clear
 - absence of visible particulate matter (sediment)
- Bright
 - absence of free water / moisture
 - no discernable haziness / cloudiness

BRIGHT?

FREE WATER ?

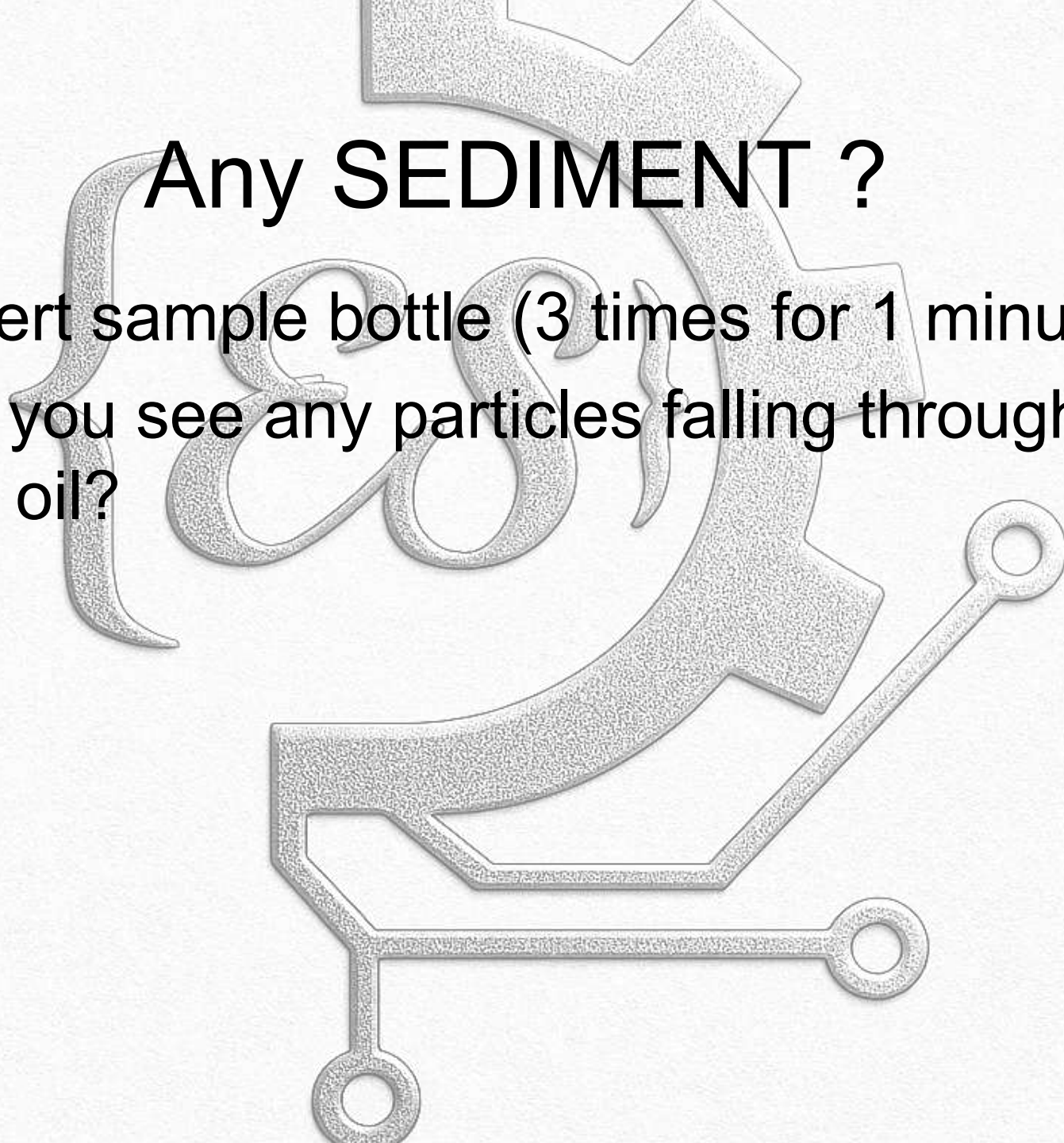
- Is there a presence of Bubbles? If there is, does it fall to the bottom (water) or rise to the top (air).
- Is there a layer of water on the bottom of sample bottle? (duh, right!)

HAZY ?

- Check sample Temp for 120 deg F
 - dissolved air/water present ?
- Wait 30 minutes
- If dissolved air
 - clear from bottom -> up
- If dissolved water
 - clear from top -> down / remain cloudy

Any SEDIMENT ?

- Invert sample bottle (3 times for 1 minute)
- Do you see any particles falling through the oil?



Transparency Test

- PMS card - Can it be read through the Sample ?
 - YES ? - Sample is BRIGHT
 - No ? - Conduct BS&W

Visible Sediment Test

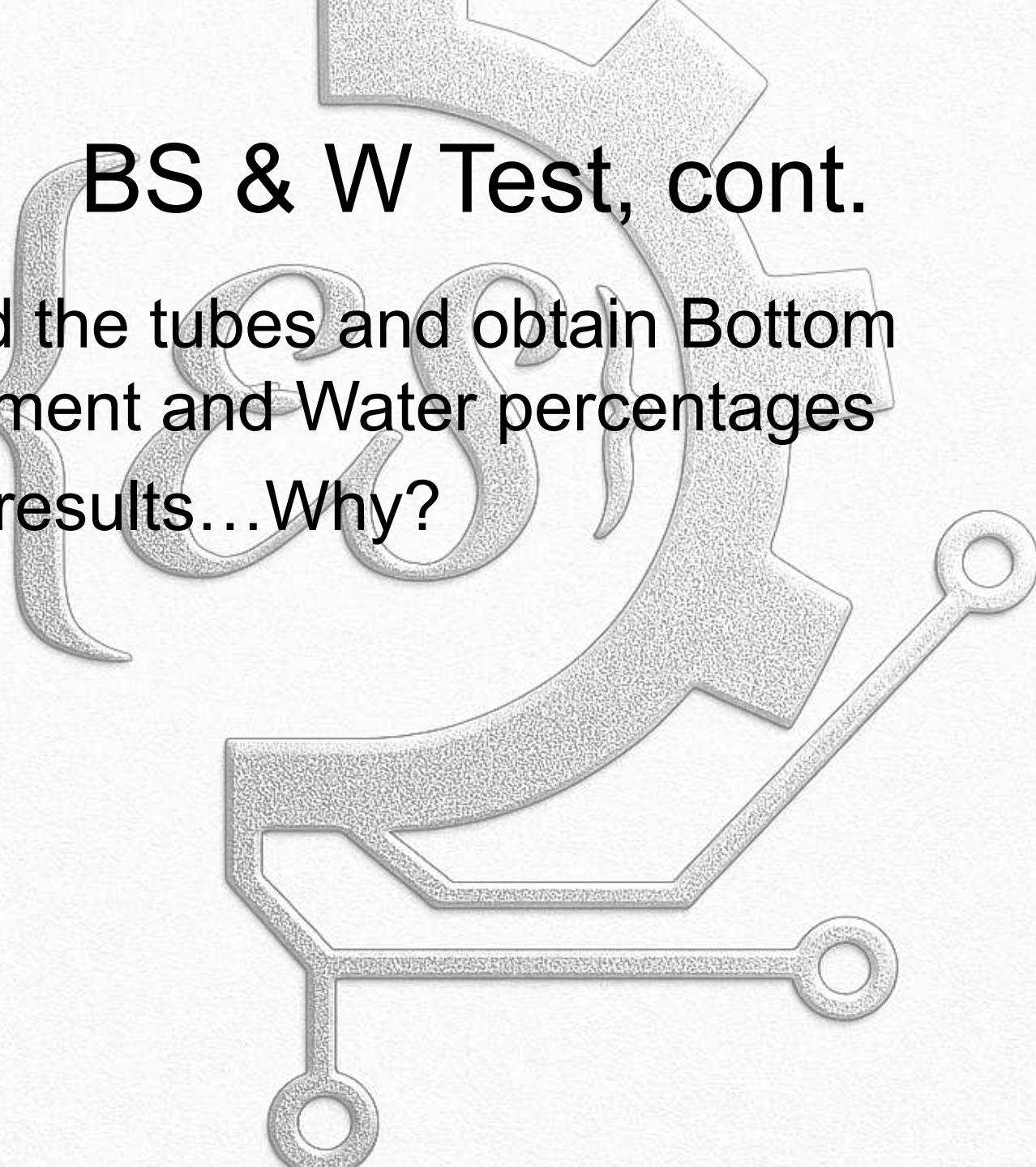
- 10 minute settle
- 10 minute “lay it down on its side”
 - Solid, Unbroken line ?
 - Individual particle $> 1/8$?
 - IF EITHER, Perform BS&W
 - IF NEITHER, Sample is CLEAR

BS & W Test

- 2 Centrifuge Tubes filled with 50ml of oil sample
- Fill tubes to the 100ml mark with solvent (P-D-680 Type II, or BIOTEK Hi-Solv)
- Cork (with Cellophane)
- Spin - 1500 RPM for 30 minutes

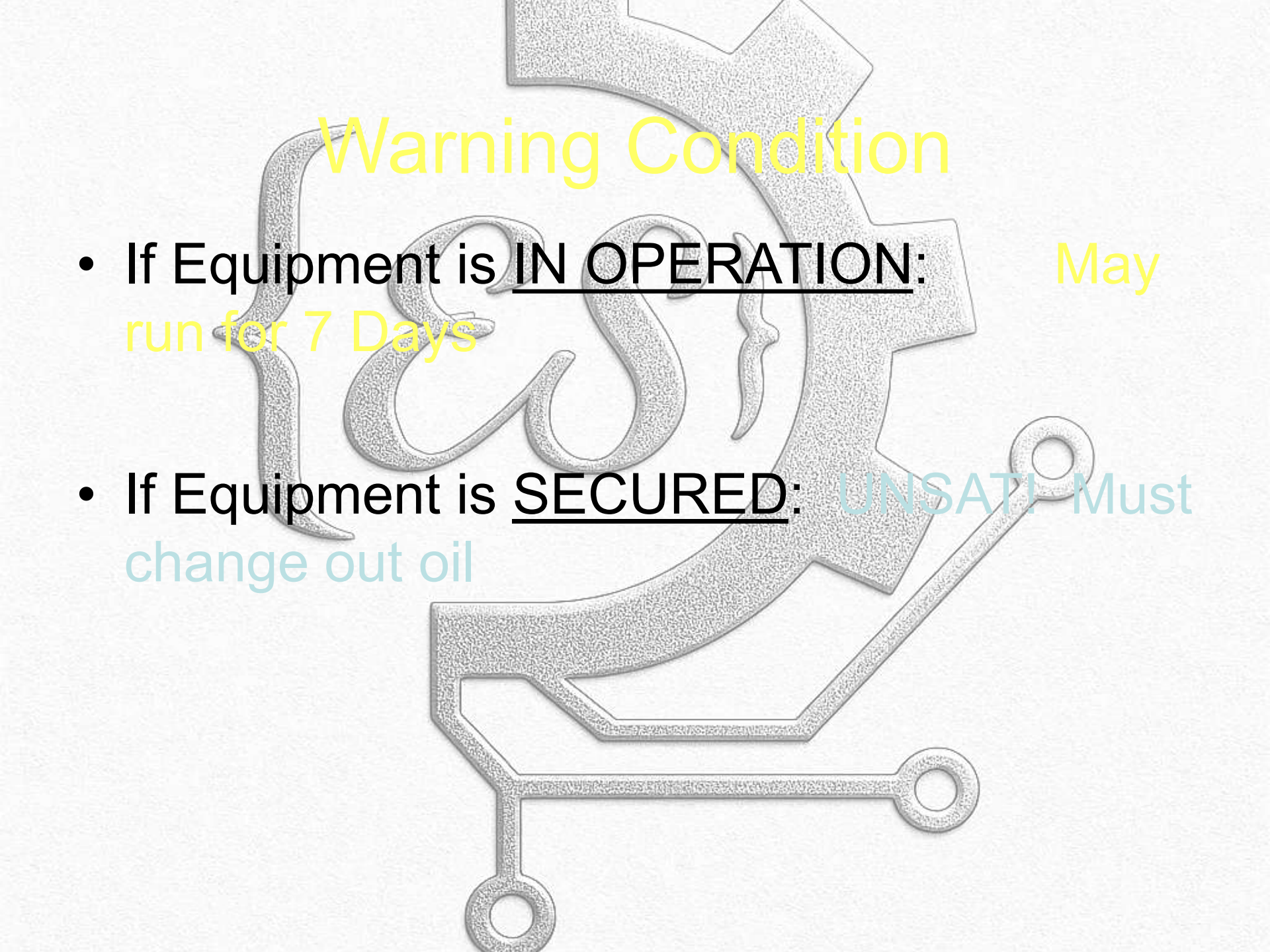
BS & W Test, cont.

- Read the tubes and obtain Bottom Sediment and Water percentages
- Add results...Why?



BS & W Results

- Sed < 0.1%, Comb. BS&W < 0.2%, **SAT SAMPLE**
- Sed < 0.1%, Comb. BS&W < 0.4%, **Warning Condition**
- Sed < 0.1%, Comb. BS&W > 0.4%, **Condemning Limits - UNSAT**



Warning Condition

- If Equipment is IN OPERATION: May run for 7 Days
- If Equipment is SECURED: ~~UNSAT!~~ Must change out oil



LOGS

Lube Oil Quality Management Log

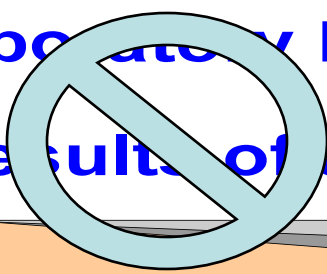
1. LO Quality Section

2. LO Sounding Section

3. LO Transfer Section

Oil Laboratory Log --

- Results of all LO tests





Log Sample

Daily LO Quality Section

DAILY LUBE OIL QUALITY MANAGEMENT LOG
 USS _____ DATE _____
ENGINE ROOM
LUBE OIL MANAGEMENT LOG
(2000 SERIES)

EQUIPMENT	SUMP CAPACITY	SUMP LEVEL	TIME OF SAMPLE	CODE	V I S U A L	S E D	T R A N S	B S & W	ACTION TAKEN	EOOW INITIALS
NR1 MAIN ENGINE										
NR2 MAIN ENGINE										
NR1 SSTG										
NR2 SSTG										
NR4 FFP										

*Sample Only
 Awaiting revision 4
 to NSTM 262*

CODES:

D - DAILY

PTS - PRIOR TO STARTUP

PTB - PRIOR TO BECCES

W

LOP

AB

- WEEKLY

- LUBE OIL PURIFIER EFFICIENCY

- AFTER BECCES

AOL

OTH

- AFTER ON LINE

- OTHER

LO Sounding/Transfer Section

LUBE OIL DAILY SOUNDING SECTION

EQUIPMENT	CAPACITY	SOUNDINGS
STORAGE TANKS		
SETTLING TANKS		
MAIN ENGINES		

LUBE OIL TRANSFER SECTION

FROM	TO	SAMPLE RESULTS	TIME	AMOUNT TRANS

REVIEWED BY: _____ MPA
LUBE OIL QUALITY MANAGER

Oil Lab Log (BS&W Log)

OIL LABORATORY LOG

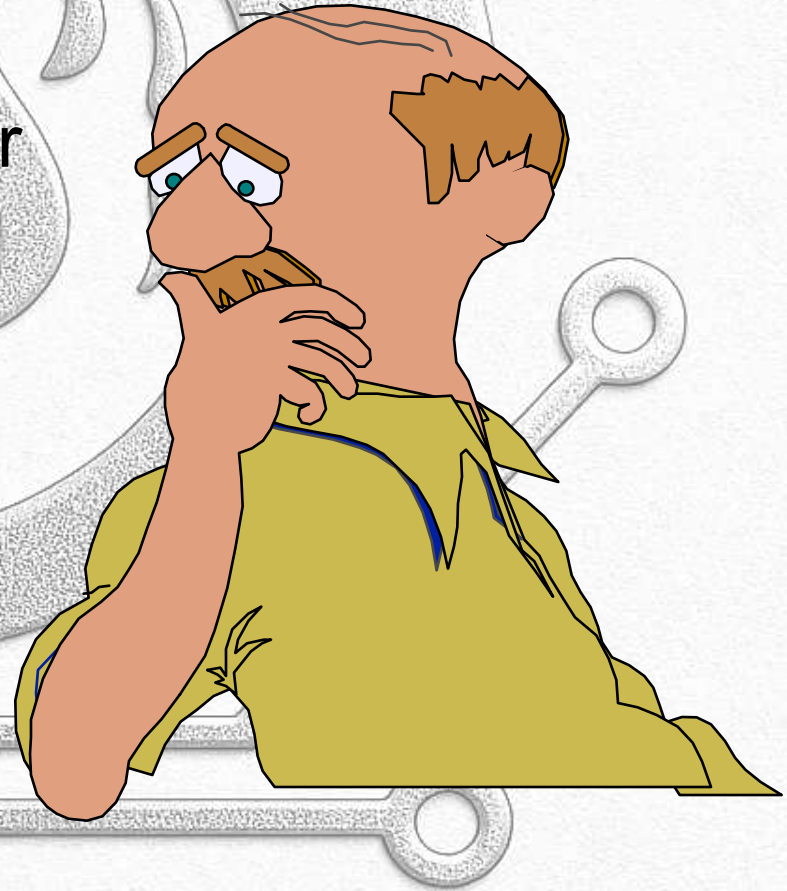
USS _____ DATE _____

DATE	TIME	EQUIP	TYPE OF TEST	TEST RESULTS	ACTION TAKEN	OIL KING INITIAL	LOQM INITIAL	MPA INITIALS

NOTE: ALL UNSAT TEST RESULTS SHALL BE CIRCLED IN RED AND INITIALED BY THE ENGINEER OFFICER.

Responsibilities

- Engineer Officer
 - Establish and Monitor Program
 - Determine if Equipment is OOC



Responsibilities (Cont.)

- LO Quality Program Manager
 - Typically MPA
 - Supervises Receipt and Strike Down of New Oil
 - Review All Lube Oil Logs
 - Establish High, Low, and Normal Sump Levels
 - Submit JOAP/NOAP Samples

Responsibilities (Cont.)

- Division Officer
 - Monitor Lube Oil Daily
- Space Supervisors
 - Sample Equip. IAW PMS and NSTMs
- Oil King
 - Perform, Record, Report, and Recommend

Naval Oil Analysis Program

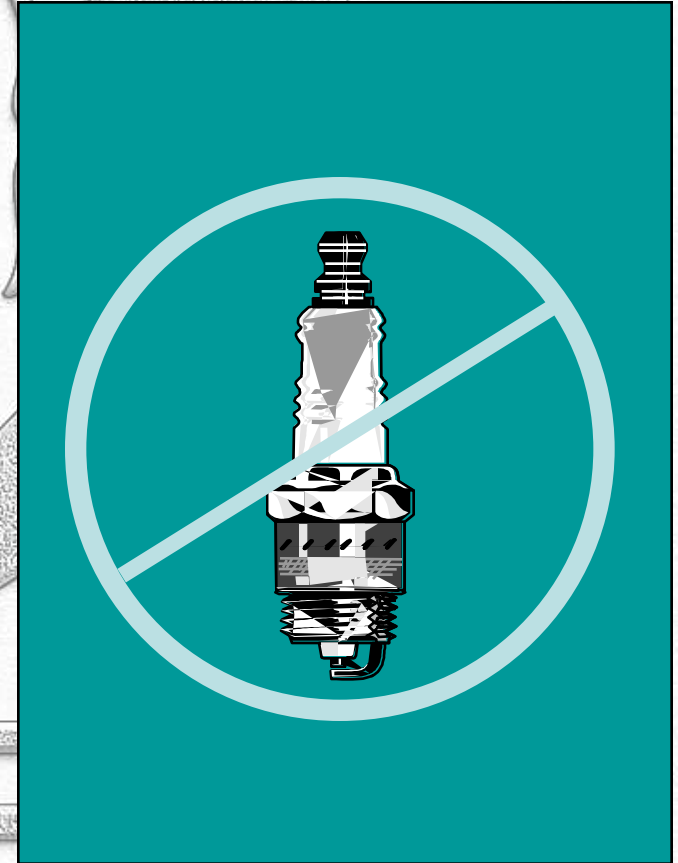
- Established to routinely monitor metal wear
- Provides trend analysis
- Spectrographic analysis of sample
- True indication of abnormal wear
- Reports/Messages must be maintained for 2 years

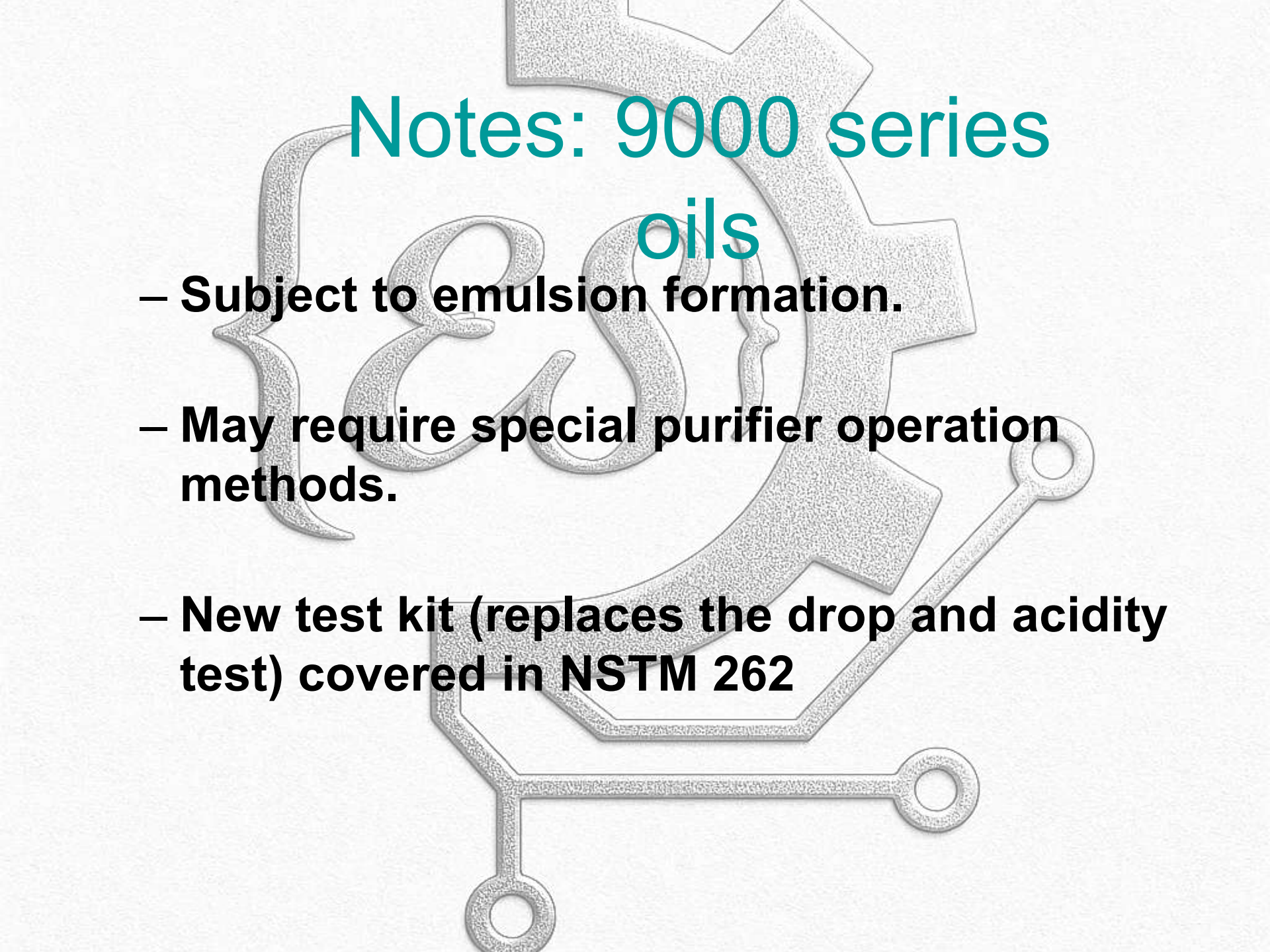
Naval Oil Analysis Program

- **95% reduction in the required PMS sample requirement, reflected in latest Force revision**
- **Extra work with no value added**
- **Remaining items: Propulsion Diesels, SSDG, EDG, CRPP, Aircraft Elevators**
- **Labs still available as needed for evaluation**

Diesel Engine LO - 9250

- Sample & Test IAW PMS and NSTM 233 / 262
- Sample Criteria
 - No “Clear & Bright”
 - Test For Viscosity & Acidity

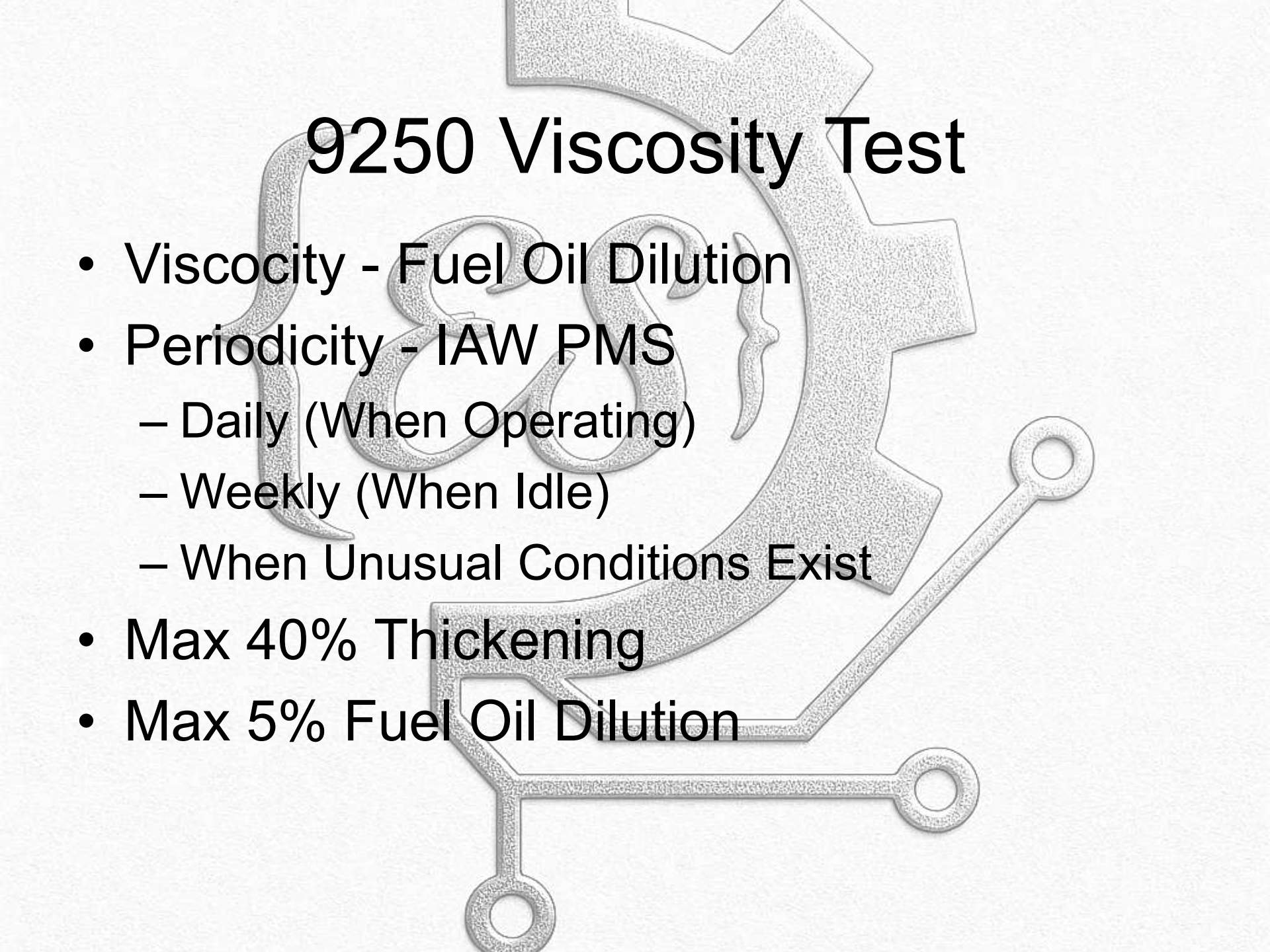




Notes: 9000 series oils

- **Subject to emulsion formation.**
- **May require special purifier operation methods.**
- **New test kit (replaces the drop and acidity test) covered in NSTM 262**

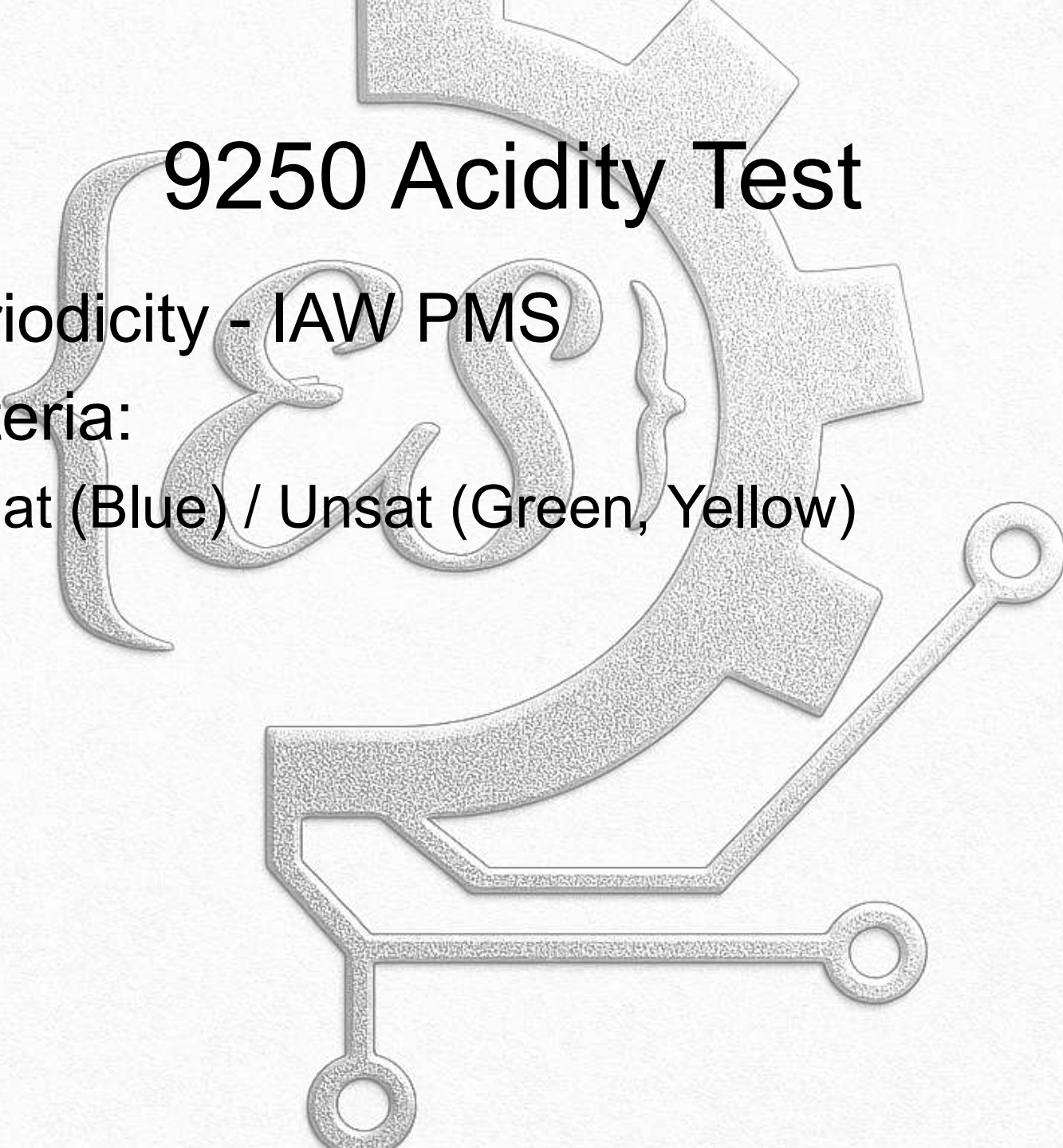
9250 Viscosity Test

A large, light gray gear is positioned in the upper half of the slide. Below it, a mechanical linkage with three circular joints is visible, extending from the right side towards the bottom center.

- Viscosity - Fuel Oil Dilution
- Periodicity - IAW PMS
 - Daily (When Operating)
 - Weekly (When Idle)
 - When Unusual Conditions Exist
- Max 40% Thickening
- Max 5% Fuel Oil Dilution

9250 Acidity Test

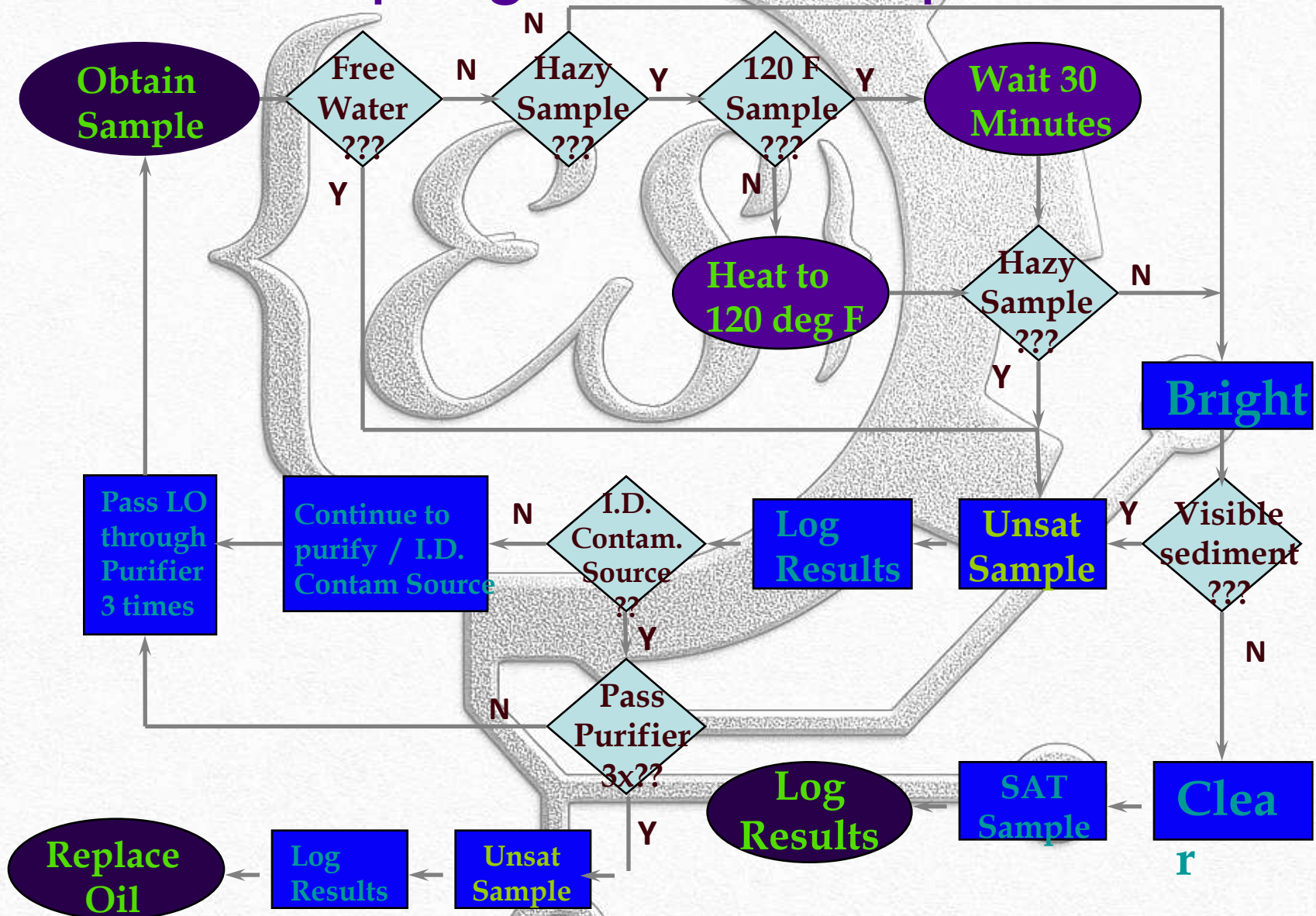
- Periodicity - IAW PMS
- Criteria:
 - Sat (Blue) / Unsat (Green, Yellow)



The Daily Lube Oil Log

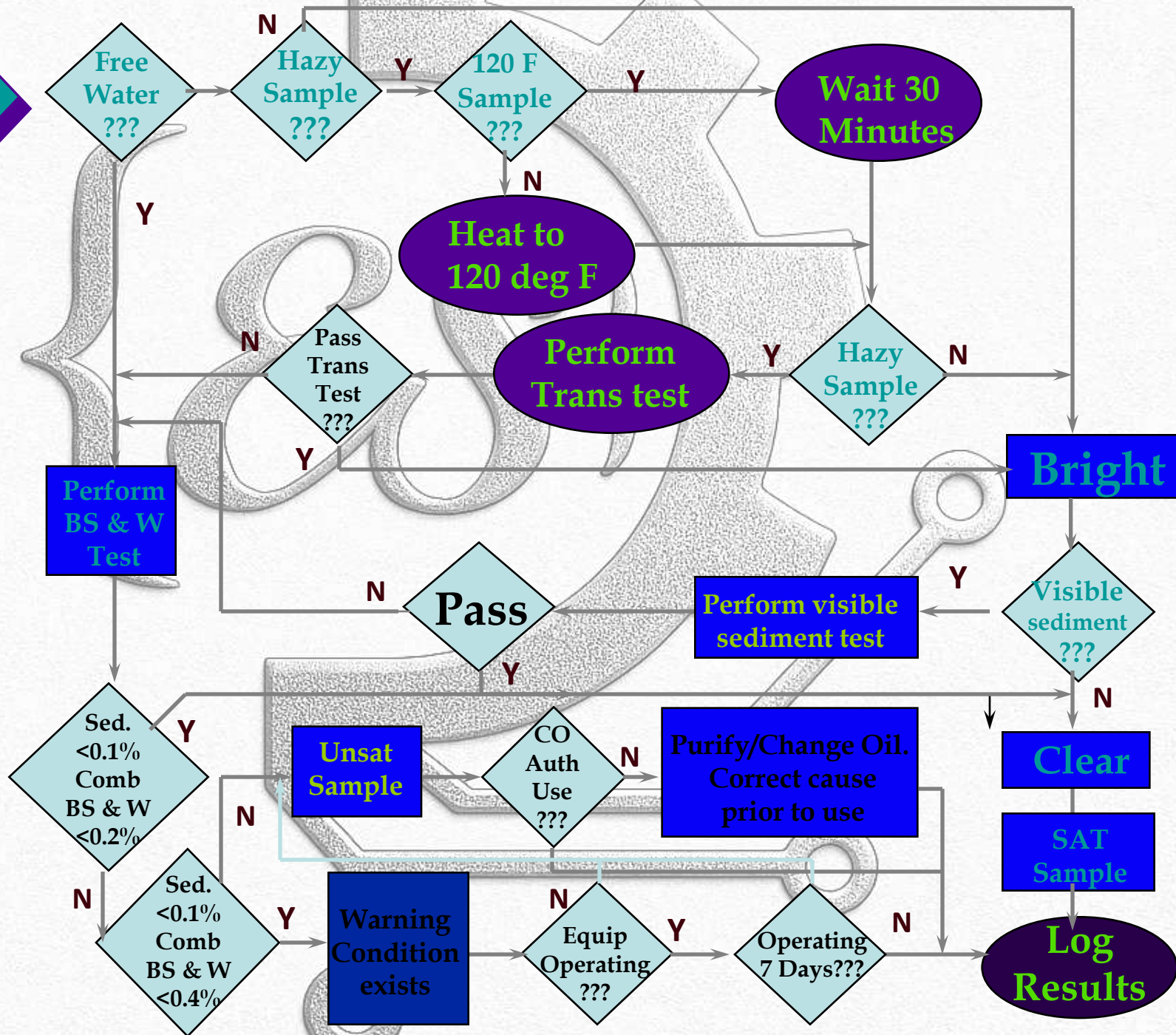
- Maintained by the space supervisor and reviewed by the EOOW/EDO.
- Record of all sampling, sounding and transfer of lube oil and the results of all lube oil testing.
- A log must be maintained for each diesel engine.
- The space supervisor and EOOW/EDO must review and initial the log for each sample.
- Unsat sample entries are to be circled and explained with the reason for unsatisfactory sample and corrective action taken.

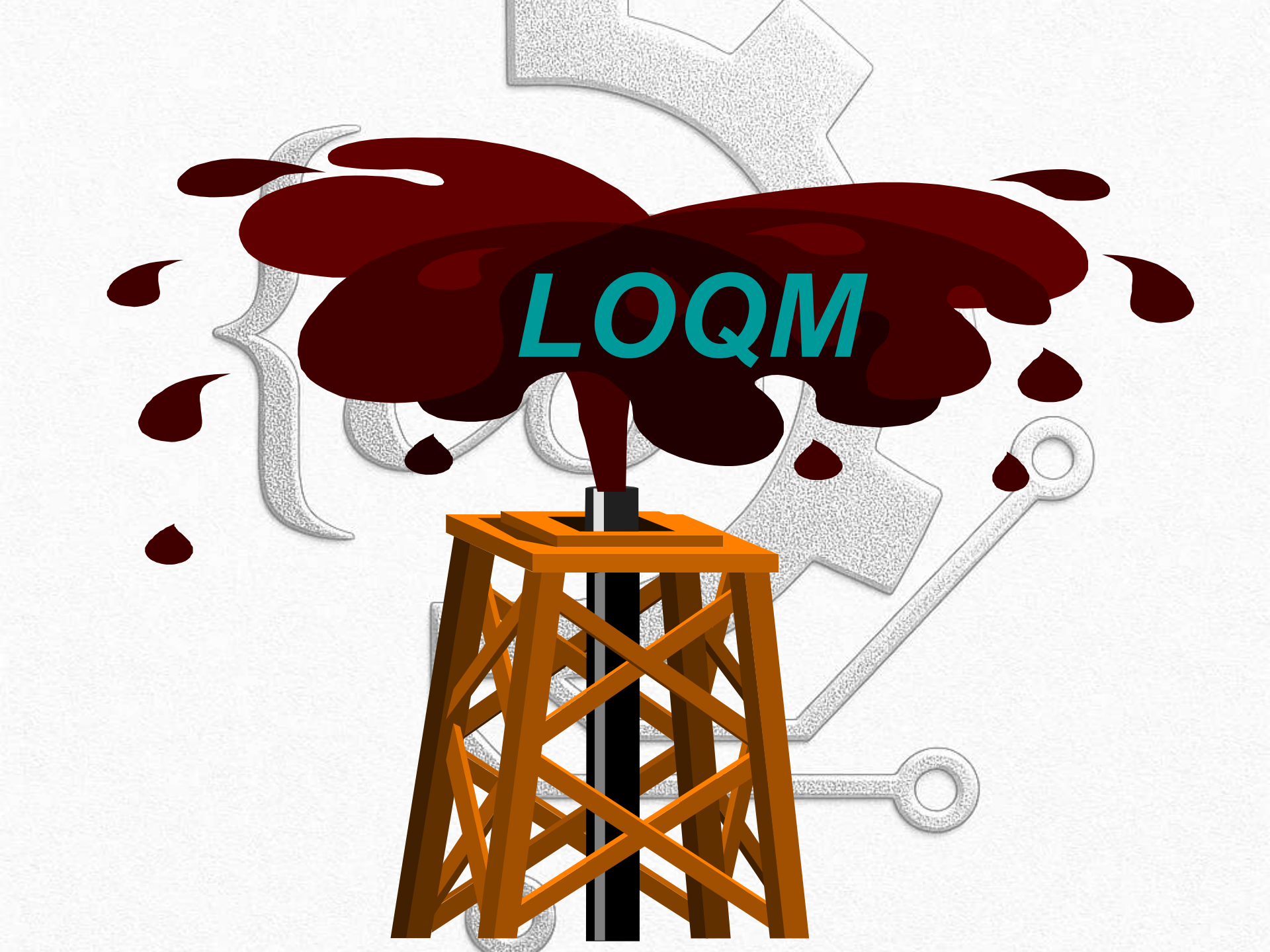
sampling with online purification



**Obtain
Sample**

*Systems without online
Purification Capability*





LOQM